

***The 1-D Analyst[®] II
Data Analysis Software
for the Model 620 Densitometer***

Catalog Number 165-2072
Version 3.0

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(b) The Purchaser must notify Bio-Rad in writing, within thirty (30) days of delivery of the software to the Purchaser (not including delivery of any subsequent modifications to the software), of its claim of any defect. If the software is found to be defective by Bio-Rad, Bio-Rad's sole obligation under this warranty is to remedy such defect in a manner consistent with Bio-Rad's regular business practices which shall mean that for a defect which materially adversely affects the performance of the Software, Bio-Rad shall use its best efforts to cure such defect as soon as reasonably practicable after receipt of Purchaser's notice, and that for minor defects, Bio-Rad shall use its best efforts to correct such minor defects in the next release of its software; provided, however, that if Bio-Rad is unable to cure a major defect within (90) days of receipt of Purchaser's notice, Purchaser shall have the option to cancel this Agreement whereupon Bio-Rad shall refund only the Software fees paid hereunder.

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Table of Contents

Section 1 1-D Analyst II Software	4
1.1 Introduction	4
1.2 System Requirements	5
1.3 System Setup	5
1.4 Notational Conventions	6
1.5 Installing Microsoft® Windows® Software	6
1.6 Installing the 1-D Analyst II Software	7
1.7 Starting the 1-D Analyst II Program	8
1.8 Working in the Windows Environment	9
 Section 2 Learning to use the 1-D Analyst II Software	 13
Lesson 1 Identification and Integration of Peaks	14
Lesson 2 Molecular Weight Determination	18
Lesson 3 Importing Data from the Densitometer	25
Tutorial Summary	28
 Section 3 1-D Analyst II Software, Description of Functions	 31
3.1 Capture	31
3.2 Display	37
3.3 Weight	58
3.4 Report	61
 Appendices	 64
Appendix A Technical Support	64
Appendix B Cable Pin-Out Tables	64
Appendix C Data Files and Data Base Management	65
Appendix D Use of Molecular Weight Calculations	65
Appendix E Trouble Shooting	66
 Index	 68

Section 1

Introducing the 1-D Analyst II Software

1.1 Introduction

The 1-D Analyst II Data Analysis Software is a powerful analytical tool designed to analyze one-dimensional scan data digitized by the Bio-Rad Model 620 Densitometer. The 1-D Analyst II Software enables analysis of virtually any one-dimensional application including: SDS-PAGE gels, IEF gels, immuno-blots, autoradiograms. The determination of valuable information such as; sample concentration or amount of macromolecule, relative mobility and molecular weight, is easily performed with the 1-D Analyst II graphical interface. Operating under Microsoft Windows® software, the 1-D Analyst II software offers pull-down menus and dialog boxes which make it surprisingly easy to learn and simple to use.

It is recommended that the first-time user follow the tutorial section of this manual to become acquainted with some of the 1-D Analyst II functions and features. More detailed information about each of the functions of the software is contained in the section for that function.

The 1-D Analyst II software offers the following features:

- The one-dimensional data from the densitometer is acquired and automatically stored to hard disk
- Up to four scans can be displayed at once, either overlaid or above one another
- High frequency noise may be filtered from data using a weighted average smoothing feature
- Any section of the scan can be zoomed as much as desired using the variable zoom tool
- Origin and front of sample can be identified for relative mobility calculations
- Area for any section under the scan curve can be rapidly determined by simply dragging the integration tool along the region(s) of interest
- All peaks of user-defined minimum size can be automatically identified
- Peaks of interest can be manually identified and/or edited
- View peak and scan information directly on screen
- Molecular weight standard curve generation with scatter plot for best fit determination

- Molecular Weight values can be determined by just moving the cursor across the unknown scan
- Scan graph can be printed at any point in the analysis
- Reports include densitometer settings, scan information, peak data, and molecular weight data

1.2 System Requirements

Equipment

Densitometer

- Model 620 CCD Densitometer
- Densitometer interference filters and accessories as appropriate for your application
- Densitometer to IBM-AT compatible interface cable

Computer

-80286 or 80386 IBM-AT compatible microcomputer (a math co-processor is strongly recommended for 80286 computers)

Additional requirements:

- EGA graphics board and monitor minimum
- 640 K of RAM minimum (2 Mb recommended)
- Minimum of 20 megabyte hard disk drive (40 megabytes or better recommended)
- two serial I/O boards with ports configured as COM1 and COM2 (only necessary if using a serial mouse)
- mouse (Logitech 3-button serial mouse is recommended)
- one parallel communications port
- any graphics quality printer supported by Microsoft Windows software

Software

- DOS operating system, version 3.3 or later
- Microsoft Windows, version 2.1 or later
- 1-D Analyst II software package, version 3.0 from Bio-Rad

1.3 System Setup

Connecting the Mouse

If you are using a serial mouse, such as the Logitech C-9 serial mouse, it can be connected to either communications port number one (COM1) or number two (COM 2) as Microsoft Windows will locate the mouse in either port. Communications port 2 (COM2) is recommended. If you are using a bus mouse, please consult the manual accompanying the mouse for installation procedures.

Connecting the Densitometer

The densitometer should normally be attached to the PC serial port configured for COM1. Connect the densitometer to the computer in one of the following ways, depending on the type of serial communications port on your computer:

- If you have a 9-pin serial communications port on your PC, you should have a cable with one 9-pin connector and one 25-pin connector from Bio-Rad, catalog number 165-2057. Plug the 25-pin connector into the RS-232 port on the densitometer and plug the 9-pin connector into the serial port of your computer.
- If you have a 25-pin male asynchronous port on your PC, attach the 9 pin to 25 pin adaptor to the 9 pin end of the interface cable. Plug one connector into the RS-232 port of the densitometer and the other connector into the asynchronous port of the PC.

See Appendix B for specific pin-out requirements if constructing your own cable.

1.4 Notational Conventions

Keyboard

Throughout this manual, characters enclosed in square brackets [] stand for a key on the keyboard. For example, [ENTER] designates that the large key labeled **Enter** be pressed. Whenever you are supposed to type something at the keyboard, the instructions will say: "Type: ". For example, Type: 1D [ENTER] instructs you to type in the characters 1D and then press the **Enter** key.

Mouse

Unless otherwise specified, all "clicking" using the mouse is performed with the left mouse button (LMB). The center mouse button, if present is not functional with the 1-D Analyst II. The right mouse button (RMB) is only functional when the mouse cross hair is on top of a graph containing an active trace.

1.5 Installing Microsoft Windows

The 1-D Analyst II software is a Microsoft Windows application, and therefore requires the complete Microsoft Windows software package for either 286 or 386 computers. Included with each Microsoft Windows software package is a word processor and a paint program. Microsoft Windows for 386 computers offers a true multi-tasking environment. Thus, with Windows software you can work with several different applications at once and switch from one to another without quitting any of them.

For detailed information on Microsoft Windows, consult your Microsoft User's Guide or call Microsoft's technical service.

INSTALLATION

To install Microsoft Windows on your hard disk, follow the procedure below.

- 1) Turn on the computer.
- 2) Enter date and time if requested.
- 3) Insert the Windows Setup disk in drive A and close the door.
- 4) Type: `a:` [ENTER]
- 5) Type: `setup` [ENTER]

The Windows SETUP program will prompt you for information and instruct you to insert diskettes one by one. If you make a selection mistake when running Setup, simply start Setup again and repeat the procedure. Refer to the MS-Windows Users Guide if you need help.

1.6 Installing the 1-D Analyst II software

The 1-D Analyst II software is supplied by Bio-Rad on both a 5 1/4 inch high density disk and a 3 1/2 inch high density disk. The contents of the two disks are identical. Select the disk that matches your computer system.

- 1) Backing up the 1-D Analyst II Program disks.
Before beginning the installation procedures, protect your investment by backing up the 1-D Analyst II Program disks.
 - Turn on and boot up your computer
 - Format a high density disk according to the instructions in your computer owner's manual.
 - Copy the 1-D Analyst II disk to the newly formatted high density disk. (The 1-D Analyst II Software is not copy protected)
- 2) To install the 1-D Analyst II software onto your computer's hard disk, simply insert the appropriate size disk into your computer's floppy disk drive, and close the latch.

Type: `A:/INSTALL` [ENTER]

- 3) The Installation program will automatically determine whether your computer has a math co-processor installed. (All aspects of the 1-D Analyst II software will operate significantly faster on computers equipped with math co-processors.) The installation program will then direct you to enter the full path name of the directory, including the drive, in which you would like to install the 1-D Analyst II program files.

If you wish to install the files in the computer's root directory type: `C:\` or `D:\` [ENTER]

If you wish to install the files into an existing sub-directory type: `C:\(filename)` [ENTER], e.g., `C:\1DANALST` [ENTER].

If you enter a sub-directory name which does not already exist, the program will give you an error message which states: "Invalid directory - Enter CTRL-C to terminate". If you wish to create a sub-directory now, press [CTRL]-[C]. To create a sub-directory type the following: *MD\sub-directory* [ENTER], e.g., *MD\1DANALST* [ENTER]. Now that you have created a sub-directory you may re-execute the Installation program by typing: *A:\INSTALL* [ENTER], and follow the instructions outlined above for load the 1-D Analyst II program files into the directory you have just created.

The installation program will then load the appropriate 1-D Analyst II executable file, according to whether or not your computer contains a math co-processor, and the Tutorial files into the drive and directory which you have specified.

1.7 Starting the 1-D Analyst II Program

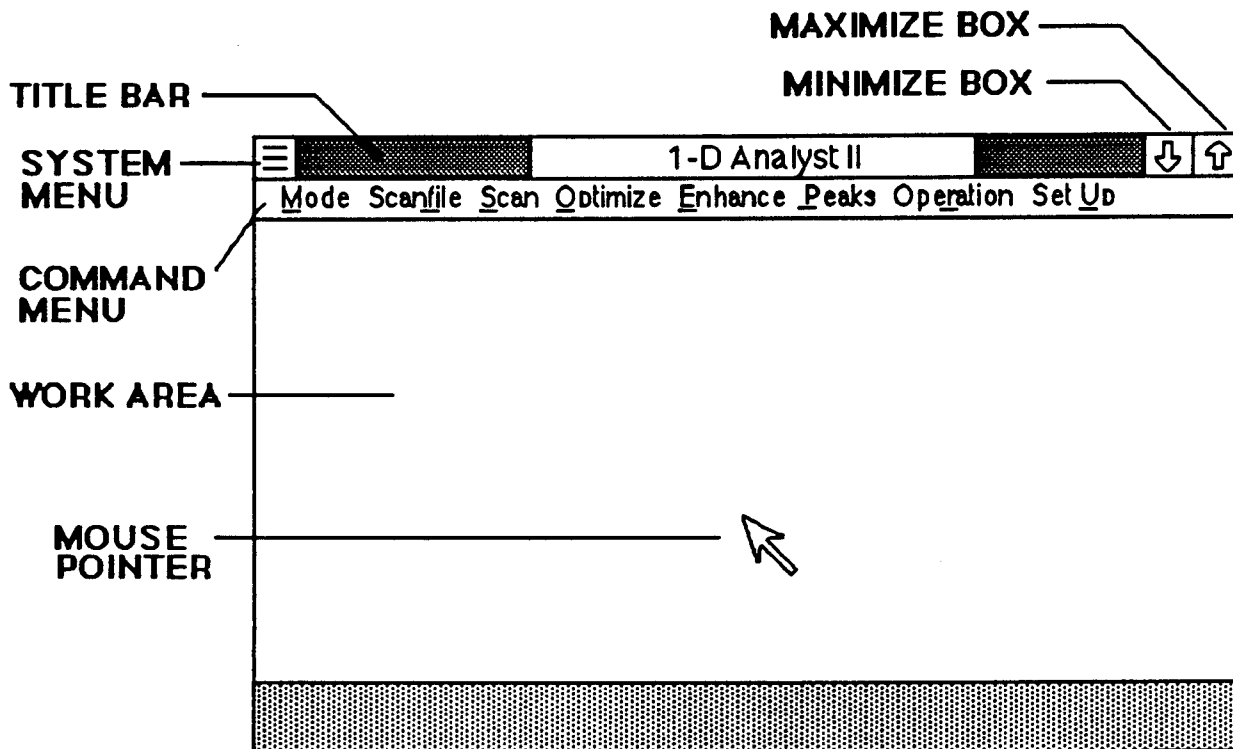
This procedure assumes that you have already installed Microsoft Windows as well as the 1-D Analyst II software on your hard disk as described in the previous section. To start the 1-D Analyst II program follow the steps below:

- 1) Type: *CD* [ENTER] or *CD\sub-directory* [ENTER], e.g. *CD\1DANALST* [ENTER], depending upon whether you installed the program in the root or in a sub-directory.
- 2) Type: *WIN* [ENTER] or *WIN386* [ENTER], depending upon whether you installed Microsoft Windows 286 or 386 in section 1.5. The screen will clear and Microsoft Windows software will load, and then display the MS-DOS Executive window. At this point the mouse pointer should be active. If the mouse pointer is not active please refer to the trouble shooting section, Appendix E.
- 3) With the mouse, move the mouse pointer until it points to the 1DA.EXE file (1DI.EXE if your computer has a math co-processor), and double click with the left mouse button. The 1-D Analyst II program will then load, displaying the title screen for a few seconds before showing the main menu.

1.8 Working In the Windows Environment

This section will acquaint you with the basic elements and operations of the program, such as using the mouse, selecting items from the command menus, and operating the dialog boxes.

The components of a 1-D Analyst II window



The main components of a window:

- The **menu bar** contains the names of the command menus (i.e., *Functions, File, Colors, Graphics, Extract, Run*)
- The **system menu box** contains a list of commands that are applicable to the system window.
- The **command menus** contain a list of commands. Commands which are valid (i.e., may be carried out) are in regular type. Commands displayed in light type are not valid, or are the default command currently in operation and can not be executed. Selecting invalid commands produces no effect.
- The **work area** is the large field in the center of the window where command information and peak reports are displayed.
- The **mouse pointer** is a small graphic symbol which appears on the screen and is used to indicate or point at the menu or function you wish to access. The mouse pointer can take the form of an arrow or cross hair, but will change shape during certain tasks. E.g., the mouse pointer takes the form of an I-beam when it is placed in a text box, and will appear as an hour glass when the program is waiting for the computer to process information.
- The **maximize box** is used to enlarge your window.
- The **minimize box** is used to shrink your window to an icon.

- **Scroll bars** appear in windows having more information than can be displayed in one screen.

Using the mouse

A mouse is an alternate input device which allows you to quickly execute complex commands and operations without using the keyboard. The mouse is connected via a serial port to your PC. When a mouse has been installed on your PC and a mouse-compatible program is running, an arrow (the "mouse pointer") will appear at the center of the screen indicating that the mouse is active.

The graphical interface of Microsoft Windows is especially well suited to the use of a mouse. The mouse controls the pointer on your computer screen. Moving the mouse around will cause the mouse pointer on the computer screen to move in the same direction. Several terms used when working with a mouse are:

A. **POINTING:** Pointing refers to moving the arrow on the screen so the tip of the arrow rests over a particular function, feature, or menu on the screen. Moving the arrow is accomplished by moving the mouse around on a flat surface.

B. **CLICKING:** Clicking refers to pressing and releasing a button on the mouse while the arrow/pointer is resting over the function you wish to access. Clicking is used to select features on the screen for future operations.

C. **DOUBLE-CLICKING:** Double-clicking refers to pressing and releasing the button on the mouse, twice in succession. This is usually analogous to one click followed by pressing the ENTER key on the keyboard and is used to select a function which has been previously highlighted.

D. **PRESS:** Pressing refers to the continuous action of holding the mouse button down. Pressing on menu titles and window menu boxes will keep them pulled down until you release them. Pressing on scroll arrows causes continuous scrolling.

E. **DRAGGING:** Dragging combines the techniques of pointing, clicking, and moving the mouse. To drag a particular feature, first point to that feature, hold down the button on the mouse, and continue to move the pointer. Dragging is used to move windows, choose commands from the command menus, or bring the peak report into full view.

Note: In this manual, unless otherwise specified, all clicking is done using the left mouse button.

Using the command menus

A command menu contains a list of commands. For example, the *SCANFILE* menu in the DISPLAY screen contains a list of three commands:

Copy Scan
Delete Scan
Compact Scan File

Valid commands are in standard type. Commands displayed in a light type are not valid, or are the default command currently in operation and can not be executed. Selecting invalid commands produces no effect.

Pulling down a menu and selecting a command

The menu commands may be revealed by pointing the mouse arrow to the desired menu item and clicking once. This will reveal ("pull-down") the menu. Once a menu has been pulled down, a function within that menu may be selected by pointing and clicking on that particular function. If no function within the current menu is desired, the menu may be erased by pointing and clicking on a location outside the menu box. The menu will disappear. An alternate and sometimes faster method of selecting menu items uses the technique of dragging. First, point to an item within the menu bar and then, while holding down the button on the mouse, move the mouse downward to reveal the menu box. The selections within the menu will be highlighted successively as you move down with the mouse. When the desired function is highlighted, releasing the button on the mouse will cause that function to be selected and executed.

Changing windows

A. MOVING WINDOWS: Using the mouse, windows which do not fill the entire screen may be easily relocated on the screen. Moving a window can reveal portions of other windows that are partially or totally hidden from view. To move a particular window, point to the title bar of the window and drag the title bar to its new location. A gray outline will show the progress of the move. When you release the button on the mouse, the window will be redrawn in the new location.

B. SIZING A WINDOW: If a window does not fill the entire screen, its size may be adjusted by the user. To size (change the dimensions) of the window, first point to the border of the window you wish to size. The mouse arrow will be replaced by two opposing arrows. Drag the border of the window in the desired direction. A gray outline will show the progress of the sizing. When you release the button on the mouse, the window will be sized according to the new dimensions.

C. MINIMIZING A WINDOW (reducing to an icon): An ICON is a compact representation of a window. Because no interaction can take place with a window in this form, an icon is usually used to suspend operation of a program (without exiting the software), so that the user can interact with another program.

A window may be reduced to an icon by pointing and clicking on the downward-facing arrow in the upper right corner of the window. To restore a window from an icon back to its original size, double click on the icon.

D. MAXIMIZING A WINDOW: A window may be maximized (expanded to fill the entire screen) by pointing and clicking on the upward-pointing arrow in the upper right corner of the window.

E. SCROLLING WITHIN A WINDOW: Often a window will not fully display all the data available within a list box. This is often the case with the list boxes, which will contain more information than can fit in the window. In this case scroll bars will appear on the

bottom and/or right side of the window. This will allow you to move the contents of the Peak Report Table within the window so you can see more of it.

To scroll one line at a time, click on the arrow that points in the direction you want to see. You can scroll continuously by pressing instead of clicking the mouse button.

To scroll one window-full at a time, click in the gray area of the scroll bar. Click above the white scroll box to go up, and click below the scroll box to go down. The amount scrolled is relative to the distance from the white scroll box to where the mouse is clicked. You can scroll continuously by pressing instead of clicking the mouse button.

To scroll to any part of the Peak Report Table, drag the white scroll box to a place in the scroll bar that approximates the area you want to see.

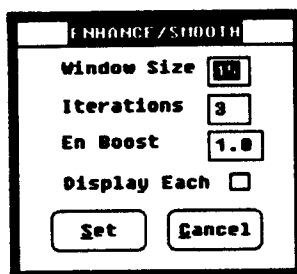
F. CLOSING A WINDOW: Windows may be closed by either selecting Close from the window command menu, or by clicking on the system menu bar.

G. ACTIVATING A WINDOW: While many windows may be open simultaneously, only one window can be active at one time. The active window is distinguished by a dark title bar. An inactive window can be made active by clicking the mouse on any visible portion of the window. Activating a window brings it to the front or top of the stack where it may partially or completely overlap other windows. To see a covered window, simply close or move the active window.

Multiple windows may be arranged on the screen to suit your needs. A window may be superimposed over other windows, or the windows may be layered so that a portion of some or all the windows are visible. Layering is a convenient method of managing multiple windows because the active window is always visible at the top of the stack. If you layer your windows so that a portion of every open window is visible, you can quickly identify any window and bring it to the top.

Using dialog boxes

Windows communicate with the user by means of "Dialog Boxes." Dialog boxes are small windows, superimposed on the currently displayed active window. They request information from the user or provide instruction or information about the current process.



(A typical dialog box presented by the 1-D Analyst II)

A. RADIO BUTTONS: The circles next to the options are called radio buttons because they are conceptually similar to the station selection buttons on a car radio. Only one of

the options from the list may be selected at any one time because the options are mutually exclusive. An option may be selected by clicking the mouse on the corresponding button. The current or default selection displays a filled-in button. When a new selection is made, the previous option is deselected, and the radio button of the new selection will appear filled-in.

B. SCROLL BAR: Scroll bars appear on list boxes on the screen. The scroll bars are used as described above to view the contents of any list that is too long to fit in the list box.

C. TEXT BOX: The text box is used to input information from the keyboard. A text box may be activated to receive information by two methods: click the mouse in the text box and a flashing cursor will appear in the box, or press the tab key until the text box becomes highlighted. Once the text box is activated, you may enter the required information from the keyboard.

D. COMMAND BUTTONS: The command buttons are used to carry out the command they display.

E. ENTRY FIELDS: The entry fields are used to accept user input of names for saving files are identifying data. The maximum number of characters that the entry boxes will accept is eight.

Section 2

Learning to use the 1-D Analyst II Software

About the Tutorial

Using the 1-D Analyst II software, you can easily and quickly determine the concentration and molecular weight of components in your samples. This tutorial introduces the skills you'll need to use the 1-D program.

The program interfaces with the Bio-Rad Model 620 Video Densitometer and with Microsoft Windows Software. Familiarity with Microsoft Windows and with densitometry procedures is assumed. If you are unfamiliar with the Windows program, please see the "About Microsoft Windows" section of this manual or the *Microsoft Windows Manual*. Please refer to the *Video Densitometer Manual* to review densitometer operation.

In Lesson 1 of the tutorial you will identify and quantitate unknown protein peaks from a scan. First, you will learn to display the scan of the unknown. Then, you will optimize the scan by subtracting background and eliminating superfluous data. And finally, you will direct 1-D to identify and integrate scan peaks.

In Lesson 2 you will find the molecular weight of an unknown. Working first with the scan of molecular weight standard proteins, you will review some of the processes introduced in Lesson 1. You'll also learn new techniques for optimizing the scan and for identifying and integrating the peaks. A standard curve will then be generated using the standard scan data, and the resulting scatter plot will be analyzed for best fit

determination. Finally, you'll determine the molecular weight of the peaks identified in the sample scan prepared in Lesson 1.

In Lesson 3 you will learn how to export data from the densitometer into the 1-D program.

Practice files, for use with these lessons, have been provided. These files contain scans of proteins, but the processes introduced can also be applied to scans of nucleic acids, with the exception of protein molecular weight determination.

LESSON 1: IDENTIFICATION AND INTEGRATION OF PEAKS

In this lesson, we will work with the scan of an unknown protein. We will, first, call the scan up for display. Then we will prepare it for analysis by subtracting the background and removing data beyond the front and origin, all of which is called "optimizing the scan." After optimizing, we will "enhance" the scan by reducing high frequency noise. Finally, we will direct the program to identify and integrate peaks of interest.

PART 1: STARTING THE PROGRAM

If Microsoft Windows software is not currently running, start the software by typing the following: win [ENTER], or win386 [ENTER], as appropriate for the version of Windows on your computer.

The Microsoft Windows title screen will be displayed briefly, followed by the MS-DOS Executive window.

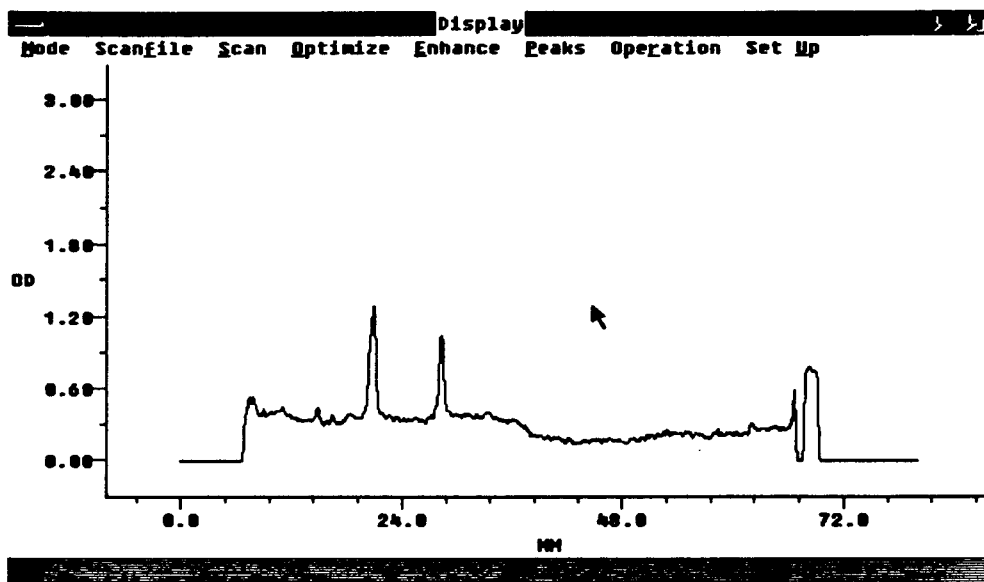
From the MS-DOS Executive window, point and click on either 1DA.EXE, or 1DI.EXE depending upon whether the 1-D Analyst II installation program installed the math co-processor version of the executable file. The 1-D Analyst II "About Box" displaying the title and software version will appear for 3 seconds, followed by the main menu screen.

PART 2: DISPLAYING A SCAN

To call up a scan for display, use the mouse to point and click on the *MODE* menu in the menu bar. Point and click on the *Display* command. The 1-D Analyst II main window will be replaced by the DISPLAY window. Point and click on the *SCAN* menu. A pull-down menu appears. Select *New* from the pull-down menu.

A dialog box appears, prompting for a filename. Point and click on the EXAMPLE.SCN file. If the EXAMPLE.SCN is not currently visible in the file box, point and click on the scrolling arrows in the dialog box until EXAMPLE.SCN is displayed. After highlighting EXAMPLE.SCN, click on the *Open* command button.

Now, a second dialog box appears, this one prompts for a scan name. Point and double click on sample.000 to open the following scan.



Note: The color bar below the scan shows x and y coordinates for the point where the cursor rests. To read the x and y coordinates for any point on the graph, simply move the cursor to that point.

PART 3: OPTIMIZING THE SCAN

To optimize the scan, we will subtract actual sample background data which was captured from the densitometer. To do so, point and click on the OPTIMIZE menu and select the *Sub Background* command from the pull-down menu. A background scan information dialog box will appear containing a list box of all of the background scans contained within the EXAMPLE.SCN scanfile. Point and click on background.000. Notice that the "Raw" and "Background" radio buttons are highlighted signifying that the scan is an unprocessed background scan. Now point and click on the **SUBTRACT** command button. The program will display the resulting scan with the background removed.

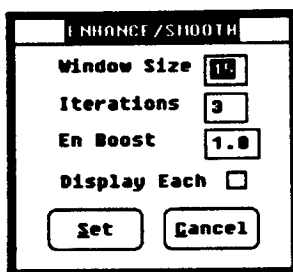
Note that the *Sub Background* command is only one of three commands that eliminate background data, each of which has advantages. The two additional background eliminating commands will be discussed in Lesson 2, part 1.

We will now trim the scan at the origin (where load sample meets resolving interface) and the front (location where dye front stopped). Click on the OPTIMIZE menu and select the *Origin* command. The cursor vanishes and a pair-of-scissors-cursor appears on screen. Using the mouse, move the scissors to the origin, where $x=6.885$. Click to clip the scan at the origin. Click on the OPTIMIZE menu and select the *Front* command. Move the scissors to the front, where $x=66.372$, and click to clip the scan at the front line.

PART 4: ENHANCING THE DATA

The scan has been optimized, and now we will reduce high frequency noise using the *Smooth* command. (Please note that 1-D also allows amplification of the spatial frequencies; information about the Enhance command can be found in Section 3.)

Prepare to *Smooth* the scan by clicking on the **ENHANCE** menu and selecting the **Criteria** command. The Enhance/Smooth Criteria dialog box appears; the cursor prompting for window size.



(In general, if you believe that your data contains a lot of noise and, therefore, needs a lot of smoothing, select a large (10-99) window size. On the other hand, if you believe that your data needs minimal smoothing, select a small (3-10) window size). Enter a window size of 15 in the Window Size edit field.

Next, tab or move with the mouse to the Iterations Value field. The iteration value specifies how many times the smoothing will be executed. For the purposes of this tutorial, enter 3.

Tab past the En/Boost edit box; En/Boost will be discussed in Section 3.

Click on the Display Each check box so that each smoothing will be displayed on screen. Set these criteria by pointing and clicking on the Set command button.

Finally, point and click on the **ENHANCE** menu and select the **Smooth** command. The scan will be smoothed and redrawn after each iteration.

PART 5: IDENTIFYING AND INTEGRATING PEAKS

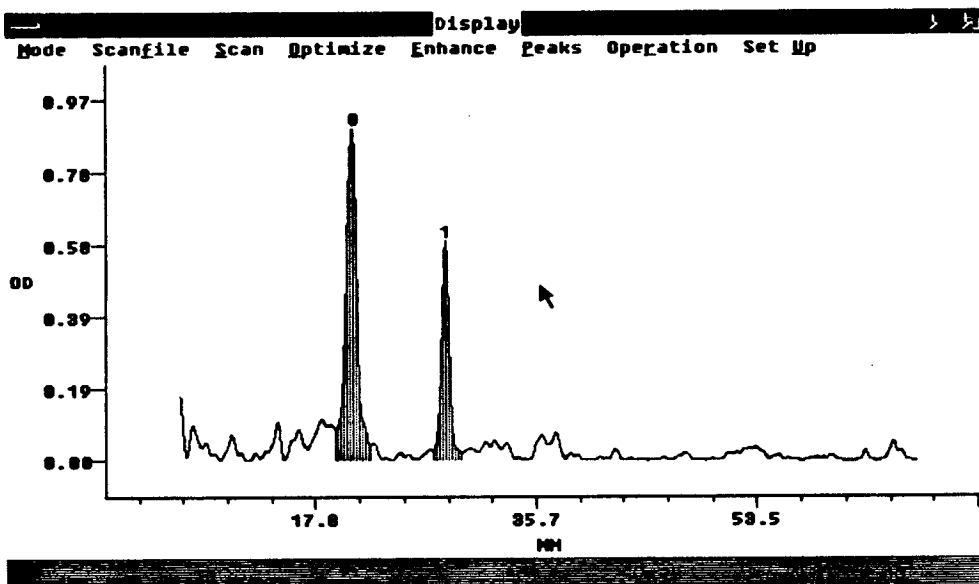
The scan has been prepared for analysis: We have eliminated background data using the **Sub Background** command; we've removed superfluous data using the **Origin** and **Front** commands; and we've reduced high frequency noise using the **Smooth** command. Preliminary work being complete, we will now identify and integrate peaks.

Begin by pointing and clicking on the **PEAKS** menu and selecting the **Peak Criteria** . . . command.

The Peak Criteria dialog box appears. In it, you establish the criteria by which you define peaks. In the Amplitude edit field enter the amplitude you consider minimum for a peak. In this case, enter 0.50. Then, using the mouse or tab key, move into the Peak Width edit field. Specify the width considered minimum for a peak. In this case, enter 0. Finally, move into the Area edit field and enter the area considered minimum for a peak; in this case, enter 0.1. To set these parameters, click on the Set command button.

PEAK CRITERIA	
Amplitude	0.50
Width	0.00
Area	0.10
Sm Win	30
Sm Iter	0
Calc'd Win	0
Set Cancel	

Now, to identify and integrate the peaks allowed by your criteria, click on the **PEAK** menu and select the **Auto Identify** command. The area under the curve for both major peaks is filled in, and drop lines identify the beginning and end of each peak. The peaks are numbered from left to right starting with peak number 0.



Note: For quick, volatile information about identified peaks go to the **PEAK** menu, and select the **Peak Information** command. A hand-shaped-cursor appears. Move the hand to the area under any peak of interest: Click. Read the peak's characteristics, including area, from the table that appears. To close the **Peak Information** box, click on the **OPERATION** menu and select the **Cancel** command.

PEAK			
Number	0		
Name			
Start	19.53	Mean	0.00
End	22.28	Std Dev	0.00
Amplitude	0.89	Amp	0.00
Amp X Loc	20.86		
Width	2.75		
Area	0.92		
Percent	35.91		
Distance	19.96		
RF	0.23		
MW			
Amount			

Or, for quick information about peaks that have not been identified, click on the PEAKS menu. Select the *Integrate Interactive* command. The cursor vanishes, replaced by an integral-sign-cursor. Move the integral-sign to the beginning of a significant peak. Press the left mouse button and drag to the end of the peak. Note that the area of the peak appears in the lower right hand corner of the screen. These area values are volatile and will disappear as soon as another peak is integrated, or the command is quit. To quit *Integrate Interactive*, click on the OPERATION menu, and select the *Cancel* command.

PART 6: SAVING THE SCAN

It is important to save after peak identification or the modifications will be lost. Therefore, to conclude Lesson 1, we will save the scan. To do so, point and click on the SCAN menu and select the *Save* command. *Save* stores the scan, with all modifications, for future use. *Save* also tags the scan name, to differentiate the scan from the original. For example, sample.000 will be saved as sample.000-a.

To close, click on the SCAN menu and select the *Close Graph* command.

Lesson 2: MOLECULAR WEIGHT DETERMINATION

In this lesson we will work, initially, with the scan of an SDS-PAGE gel lane containing standard proteins. We will review some of the techniques introduced in lesson 1, and some new commands will also be introduced. The scan will be optimized. Peaks will be identified and integrated. Then, we will associate the peaks to corresponding protein names and molecular weights and produce a log molecular weight vs. relative mobility curve for the standard proteins. Finally, we will compare the standard scan to the scan of the unknown prepared in Lesson 1. By comparing the scans we will determine the molecular weight and relative mobility of the unknown proteins.

PART 1: DISPLAYING AND OPTIMIZING THE SCAN

If 1-D is not currently running, start the program and advance to the DISPLAY window. (For help starting the program, see Lesson 1, parts 1 and 2)

In the Display window, point and click on the SCAN menu, and select the New command. When the successive dialog boxes appear, select first, the EXAMPLE.SCN file and then, the BRstandards.000 scan.

Next, prepare BRstandards.000 for analysis by eliminating data beyond the front and origin and by subtracting background.

To begin, click on the OPTIMIZE menu and select the Origin command. Use the scissors-cursor to clip the scan at $x=7.387$. Click on the OPTIMIZE menu and select the Front command. Use the scissors-cursor to clip the scan at $x=61.452$.

We will now zoom in on the section of the scan of interest using the *Zoom* command. To *Zoom*, press the right mouse button. A small, pull-down menu appears; drag to select the *Zoom* command from this menu and then release the right mouse button. Notice that the cursor has become a magnifying glass.

Practice executing the *Zoom* command by positioning the magnifying glass above and to the left of a portion of the scan that interests you. Press the left mouse button and drag to box the desired area; release the mouse button. The boxed area now appears at higher magnification. The zoom function is completely variable allowing you to zoom in as much as you wish

Now, return to the previous, unmagnified scan. To do so, press the right mouse button and select the *Previous* command from the pull-down menu that appears. Again, press the right mouse button and select *Zoom*. This time we will magnify the whole of the scan so that a baseline can be readily constructed. Position the magnifying glass cursor slightly above the highest peak, and to the left of the leftmost end of the scan. Press the left mouse button and box the entire scan. Release the mouse button to execute the *Zoom*.

By zooming we have performed a "view modification". A view modification is a modification which effects only the display, or viewing of the data, but does not actually modify the data itself. When a view modification has been done, any operation performed at this point will still effect the whole scan, not just the section which is currently in view. In order to just work with the zoomed section of the scan we must perform a "data modification". A data modification is a modification which not only effects the viewing of the data, but actually transforms the data itself. All operations performed after the data modification will only effect the portion of the scan currently in view. To work with only the zoomed portion of the graph, point and click on the OPTIMIZE menu and select *Snip at View*.

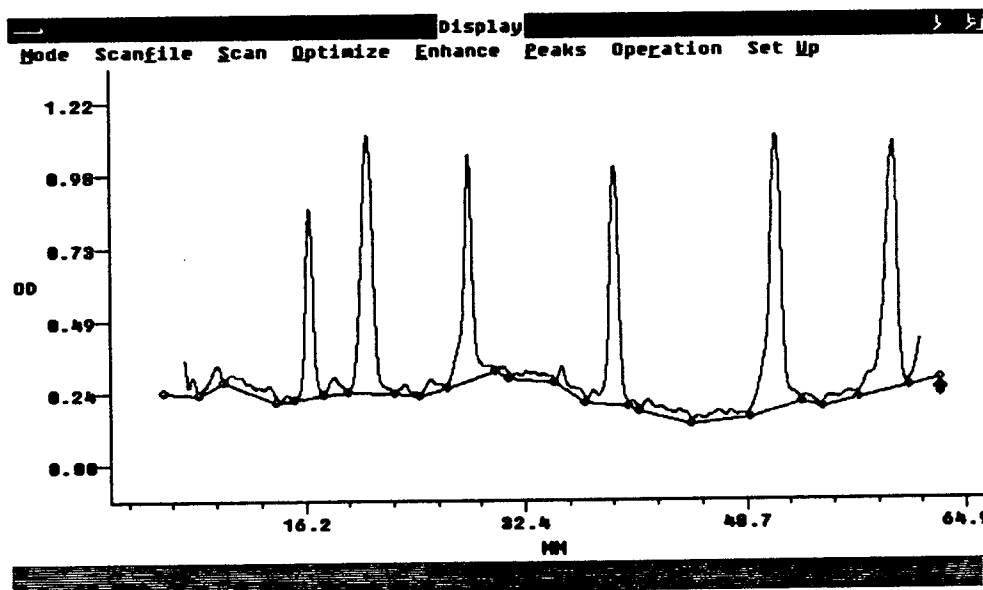
Next, we will smooth the scan using the criteria and technique outlined in part 4 of lesson 1.

Now we will subtract background data. The *Sub Background* command has been introduced. *Sub Horizontal Baseline* and *Construct Baseline* the two additional commands for eliminating background data, are presented below.

The Sub Horizontal Baseline command allows you to set a horizontal baseline and subtract it from the scan. Click on the OPTIMIZE menu and select Sub Horizontal Baseline. The cursor changes to a horizontal line extending across the graph which can be adjusted in the vertical direction. Move the horizontal baseline down the graph until the Y (O.D.) value in the lower left corner of the graph reads around 0.209. Press the left mouse button and the horizontal line will now become the new baseline (0 O.D.) eliminating all area below the line.

Now, so that the third and final command for background elimination can be introduced, click on the SCAN menu and select the Previous command to restore the unmodified scan.

The Construct Baseline command lets you draw a valley to valley baseline which is useful for scans containing a background envelope, or a drifting baseline. To use this command, point and click on the OPTIMIZE menu and select Construct Baseline. Notice that the cursor becomes a pointer. Use the mouse to move the pointer to a spot to the left of the left end of the scan and click. Then follow the contour of the scan, moving from valley to valley pointing and clicking. See the figure below for an example of baseline construction. Finally point and click on a point to the right of the right end of the scan. Be sure that you always extend the constructed baseline beyond the right and left ends of the scan. Once the baseline is constructed, point and click on the OPERATION menu. Select the Complete command to subtract the data below the baseline.



PART 2: IDENTIFYING PEAKS ON THE STANDARD SCAN

Having prepared the scan we will now integrate peaks. Begin by setting criteria for peak identification. To do so, point and click on the PEAKS menu and select the Peak Criteria command. Enter 0.10 in the Amplitude edit field. Enter 0.00 in the Width edit field. Enter 0.10 in the Area edit field. And click on the Set command button.

Then, return to the PEAK menu and select the Auto Identify command.

In some cases, you may choose to modify the peaks integrated by the *Auto Identify* command. The *Peak Edit* function lets you to add, delete, or reintegrate the peaks on your scan. To use the *Peak Edit* function, point and click on the **PEAKS** menu and drag to select *Peak Edit*. The currently displayed window closes, and the EDIT window opens.

Select **DELETE** from the menu bar; notice that the cursor becomes a hand grenade. Position the arrow of the grenade on the filled area of peak 0 and click. The peak is deleted, and filled area removed.

To adjust the width of peaks, click on **ADJUST** in the menu bar. Notice that the cursor becomes a tuning-fork. Using the mouse, move the tuning fork into the area under peak 3. Click and the tuning-fork-cursor vanishes, replaced by a wand cursor.

To adjust peak 3, position the wand cursor at 26.496 and drag it to 31.496. Release the wand and the width is adjusted. The wand will now change back to a tuning fork for adjusting more peaks. (Creating peaks using the **CREATE** command is performed in the same manner as the **ADJUST** command and will not be performed now.)

To exit the EDIT window, point at the **EDIT** menu and select the **Exit** command.

Next we will save the modified scan. To do so, click on the **SCAN** menu, and select the **Save** command.

We will conclude this part of the lesson by printing the active scan which is currently displayed. Simply press the right mouse button and select **Print**. The 1-D Analyst will automatically send the graph to the Windows spooler for printing.

PART 3: CALCULATING A STANDARD CURVE

The standard scan has been optimized, and its peaks have been identified and integrated. We will now associate standard protein names and molecular weights with the peaks within the standard scan. Then we will direct 1-D to calculate a relative mobility vs. log molecular weight standard curve.

Begin by clicking on the **MODE** menu and selecting the **WEIGHT** function. The currently displayed window vanishes and the MOLECULAR WEIGHTS window appears. Click on the **FUNCTIONS** menu and select the *Protein Editor* command. A dialog box appears; in it, we will tabulate names and molecular weights for the standard proteins. Subsequently, the table will be used to associate standard protein names and molecular weights with the peaks within the standard scan.

Protein Editor - (untitled)

Next Protein Enter

Prev Weight Delete

Entry • of •

New Open... Save... Done

Enter the names and molecular weight values for the Bio-Rad SDS-PAGE Low Molecular Weight Standards as follows:

```
ppb [TAB] 97400 [ENTER]
bsa [TAB] 66200 [ENTER]
ova [TAB] 45000 [ENTER]
ca [TAB] 31000 [ENTER]
sti [TAB] 21500 [ENTER]
lysosome [TAB] 14400 [ENTER]
```

Click on the Save command button. A dialog box appears prompting for a filename. The table will be stored in the named file. In this case, type: BRLMWSTD and click on the OK command button.

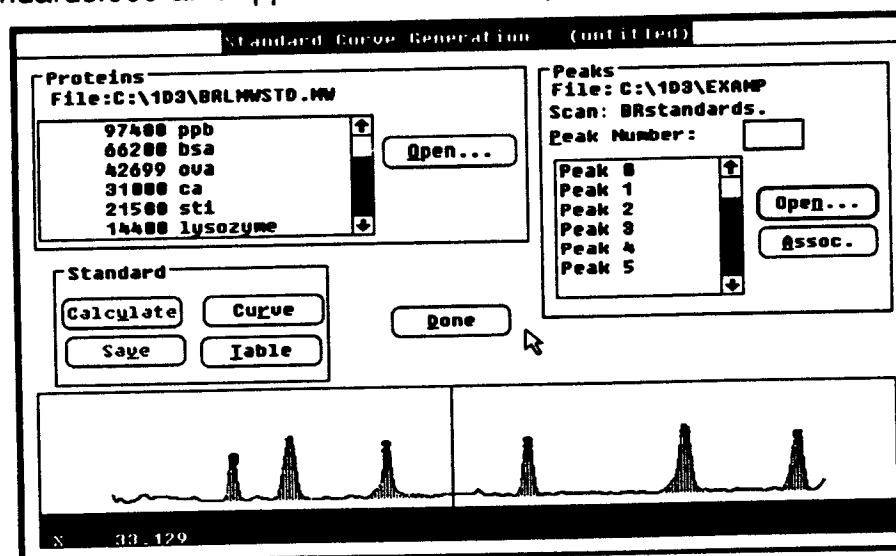
Note: tables of protein names and molecular weights for the Bio-Rad high and low molecular weight standards are provided with the 1-D program and do not have to be entered.

Now, to calculate log molecular weight vs. relative mobility curve for the standard scan, return to the FUNCTIONS menu and select Calculate Standard command.

A full screen dialog box appears with interior fields: the protein field, the peak field, and the standard field.

First, click the Open command button in the protein field. A dialog box appears, prompting for a filename. Double click on the file that holds the table of names and molecular weights for your standard proteins. In this case, select BRLMWSTD.MW. The names and molecular weights just tabulated appear in the protein field file box.

Next, click the Open command button in the peaks field. Again, a dialog box appears, this one prompting for the name of the file in which the standard proteins scan is stored. Select EXAMPLESCN. A second dialog box appears prompting for the standard scan name, select BRstandards.000-a. Click on the Add command button to open BRstandards.000-a. It appears on the lower part of the screen.



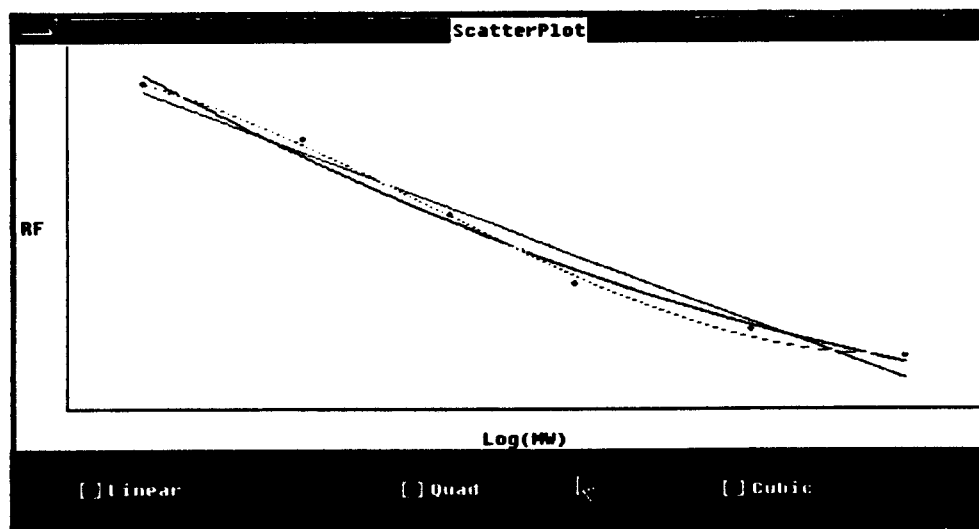
Now we will associate the protein names and molecular weights from the table with the peaks on the standard scan.

To do so, highlight both the protein listed first (ppb) and the peak listed first (0) and click on the *Associate* command button. Continue associating proteins and peaks down the tables by simply continuing to press the *Associate* command button. Note that the proteins are listed in descending order of molecular weight and that the peaks are listed in descending order of relative mobility. In this case, because there are six proteins and six peaks, a 1:1 correspondence between molecular weight and peaks is possible.

After the six proteins and peaks have been associated, we are ready to calculate a standard curve.

In the standard field, click once on the *Calculate* command button. A dialog box appears; it contains check boxes in which curve regression type may be specified. In this case, click on all three check boxes, Linear, Quadratic, and Cubic. Finally, click on the *OK* command button.

Now, click on the *Curve* command button. A scatter plot graph of relative mobility vs. log (MW) appears. So that you can evaluate the fit of each curve: click in the Linear check box to see the linear curve drawn; click in the Quadratic check box to see the quadratic curve drawn; and click in the Cubic check box to see the cubic curve drawn. To remove any of the curves, simply click again on the appropriate check box. Due to the sigmoidal shape of an SDS-PAGE standard curve, the third order regression will often give the best fit of the curve. The user is however cautioned when using higher orders of regression (Appendix D).



After you've seen all curves, close the graph by clicking once on the small menu bar in the upper left hand corner of the box. Select *Close* from the menu that appears.

We will conclude this part of the lesson by storing the curves with the standard proteins scan. To do so, click on the *Save* command button. Note: To see a list of the

proteins, relative mobilities, and molecular weights, click on the *Table* command button. To quickly close the table double click in the upper left hand corner on the menu bar.

When finished click on the *Done* command button.

PART 4: CALCULATING THE MOLECULAR WEIGHT OF AN UNKNOWN

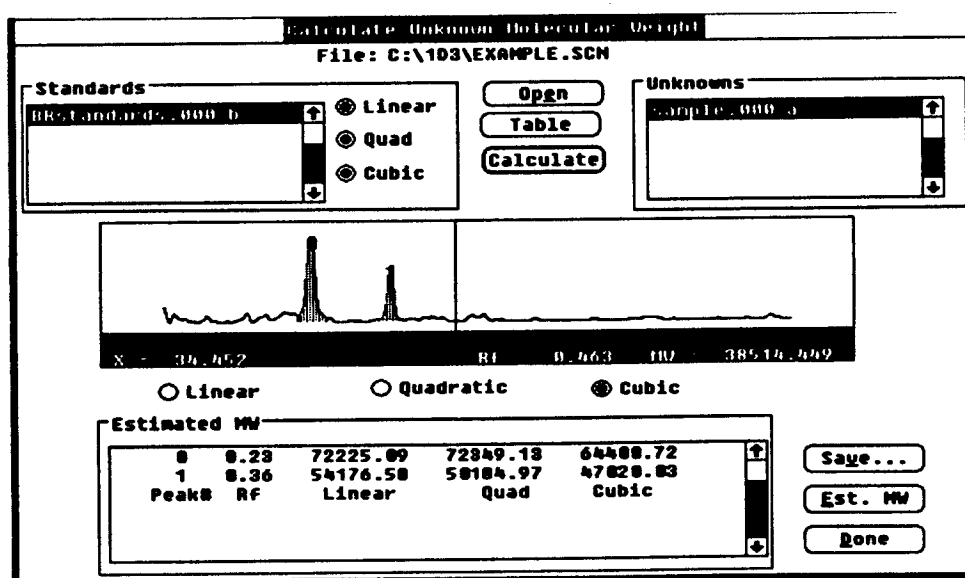
Having prepared both the scan of the standard protein and, in lesson 1, the scan of the unknown, we will now compare the two in order to find the molecular weight of the unknown proteins.

First, point to the *FUNCTIONS* menu and select the *Calculate Unknown* command. A full screen dialog box appears. Click on the *Open* command button, and double click on the EXAMPLE.SCN file. Notice that all unknown and standard scans stored in this file are listed in the appropriate file boxes.

Next, highlight the standard scan that you would like to compare to the unknown by clicking once on it. In this case, select BRstandards.000-a; recall that curves of relative mobility vs. log molecular weight have been stored with this scan. Then, highlight the unknown. In this case select sample.000-a.

Finally, click on the *Calculate* command button. The scan of the unknown appears on screen as well as three estimates of the unknown's molecular weight. Notice that the molecular weight is calculated using linear, cubic, and quadratic regressions. Move the cursor into the graph of the unknown and notice how the cursor changes to a vertical bar. As the bar is moved across the scan, the relative mobility and molecular weights values are continuously display on the bottom of the graph. By clicking on the Linear or Quadratic radio buttons below the graph, the read out will change to reflect that order of regression.

Click on the *Save* command button to save this information for later reports.



Click on the *Done* command button to exit the Calculate Unknown dialog box.

PART 5: PRINTING A REPORT

Results of your analysis can be printed as reports. You can print a report that lists the relative areas of the scan peaks and/or a report that lists the molecular weights of each component in a scanned lane.

To print, click on the MODE menu and select the Report function. The REPORTS window appears. Point and click on the FUNCTIONS menu and select the Contents command. A dialog box appears in which the type(s) of report to be printed may be specified. In this case, to see each kind of report, click in the Densitometer Data, Scan Data, Peak Report, and Molecular Weight Report check boxes. Click on the Set command button to set these choices.

Next, click again on the FUNCTIONS menu and select the Print command. A dialog box appears, prompting for a filename. Select EXAMPLE.SCN and click on the Open command button. A second dialog box appears prompting for scan name. Select sample.000-a; click once on the Add command button to begin printing reports about this scan.

To print any of the scans, such as the standard scan with the protein name above it, click on Mode and select Display. Click on the Scan menu and select New. Open the EXAMPLE.SCN file by double clicking on it. Now double click on BRstandards.000-a. Once the scan is displayed, press the right mouse button and select the *Print* command.

LESSON 3 IMPORTING DATA FROM THE DENSITOMETER

In the previous lessons, we have worked with scans provided with the tutorial. Now you will learn to transfer live data from the densitometer into the 1-D program.

PART 1: PREPARING THE VIDEO DENSITOMETER

First we will prepare the densitometer to transfer data.

To begin, set the switch on the front of the densitometer to "T" and turn the power on. The message, "Press center button to continue," will appear at the bottom of the window. Press the center button of the mouse.

Next, set the communications parameters as follows:

Choose the DIRECTORY menu.

Choose #19, OUTPUT DEFINITION.

Choose RS-232 and Register=Current; then return to the DIRECTORY menu.

Now, choose #18, RS-232 COMMUNICATIONS.

Choose Baud Rate: 19200 for an AT or clone, 9600 for XT, PC, or clone.

Choose Even Parity.

Return to the DIRECTORY menu.

Choose # 12, FILTERING AND ENHANCEMENT. Set parameters as follows: Current Setting A, Filter Frequency None, Enhancement Frequency None, Boost Factor 0. Then, return to DIRECTORY.

Choose #13, BASELINE AND PEAK SUBTRACTION. The following message should be displayed: CURRENT SETTING A, BASELINE SUBTRACT OFF.

Finally, choose # 1, Take Data Function. Choose #4, Instrument Calibration. Refer to the densitometer manual if an error message appears.

PART 2: PREPARING THE 1D-PROGRAM

Now we will prepare the 1-D program to receive data.

If 1D is not running, start the program and advance to the DISPLAY window. (For help starting the program, see Section 1.7 or Lesson 1, parts 1 and 2.)

Then Click the MODE menu and select Capture. The large, Capture dialog box appears.

Capture Scan (Untitled)

Scan

Root Name

Type

Background ☐

Standard ☐

Unknowns ☒

Densitometer Settings

Filter Freq

Enhancement Freq

Boost Factor

Baseline Subtracted ☐

Min Max

X(PM)

Y(OD)

Track Loading

Save Parameters

File

Create Append

Interpolation

☒ On

Transfer

Set Comm Capture Stop

Exit Cancel

Notes

Status

Select a File for Saving Scan Data

In the Capture dialog box, we first establish a file for saving the imported scans. To do so, click on the *Create* command button; a standard Microsoft Windows entry box appears prompting for a filename. Type in PRACTICE.SCN and click on the *OK* command button. Typically all scans from one gel will be stored in one file.

Note: a filename must be accompanied by the .SCN extension or the program will not recognize the scanfile.

Now, point and click at the Root Name edit field. Enter a name for the lane you are scanning. In this case, type: BCKGRND because we will be taking a background scan. To identify BCKGRND, click twice on the Background radio button so that it is highlighted. Notice the Standard and Unknown radio buttons; they should be selected when standard and unknown scans are captured.

Next, tab past the first three Densitometer Settings; densitometer settings will be introduced in Section 3.

Using the mouse, move to the Baseline Subtracted check box. Be sure that it is unchecked. This indicates that the software, not the densitometer, will carry out baseline subtraction.

X and Y should be at their default values: X: 0, 200 and Y: 0, 300. If x and y are at different values, type in the default settings. X and Y will be discussed in Section 3.

In the Track loading edit field, enter the quantity of sample loaded in the lane. In this case, enter 0.

Finally, click on the *Save Parameters* command button.

Next, click on the *Set Communications* command button located in the Transfer field box. Then select a Port: Com1, Baud rate: 19200 for an AT or clone, 9600 for XT, PC, or clone. Click once on the *Even Parity* radio button. And finally, click on the *OK* command button.

We are ready to capture data. Click on the *Capture* command button. Notice that the Status box reads "Output data from the densitometer."

To capture a standard or unknown scan follow the same procedure with one exception: identify standards and unknowns as such by clicking on the appropriate radio button.

PART 3: CAPTURING DATA

Having prepared the densitometer and the 1-D program, we will now transfer data. Please use the DNA film provided with the densitometer for this exercise.

Place the film on the center of the glass platen with the lanes oriented left to right. Move the platen with the left and right buttons of the mouse until the scanning line is over an area of the film that contains no sample. If you would like help, refer to the *Model 620 Video Densitometer Instruction Manual* or call Bio-Rad technical service.

On the densitometer take data screen, choose #1, Take Data/Operations. After approximately 6 seconds the densitometer will switch to the Operations screen. From the OPERATION menu. Choose Output Data.

On the 1-D screen, notice that the runner box moves from left to right and that the message "capturing data for BCKGRND.000" appears in the status box. The message "transfer complete, ready for next scan" signals a successful capture. Repeat this procedure for capturing sample data, remembering to click on the Unknown radio button and the *Save Parameters* command button before pressing *Capture*. Return to the 1-D main window by clicking on the *Exit* command button.

To verify the capture, you can view the captured scans using the *Display* command; for help with displaying, see Lesson 1, part 2.

PART 4: DELETING THE SCANS

We will close the tutorial by deleting the BCKGRND.000 scan to free computer memory.

To delete, point and click on *DISPLAY*. Click on the *SCANFILE* menu, and select the *Delete Scan . . .* command.

A dialog box appears prompting for a filename; select the file in which the scan to be deleted is stored. In this case, select PRACTICE.SCN. A second dialog box appears prompting for a scan name. Select BCKGRND.000. Click on the *Delete* command button. Click once on the *Exit* command button.

Note: Files, unlike scans, can be deleted from the MS-DOS Executive window.

Tutorial Summary

To start the 1-D program:

- Type either win [ENTER] or win 386 [ENTER]
- From the MS-DOS Executive Window, select 1DA.EXE or 1DI.EXE

To display a scan:

- From the *MODE* menu, select the *Display* command
- From the *DISPLAY* menu, select the *New* command
- In the Filename dialog box, identify the file in which the scan of interest is stored
- In the Scan name dialog box, click on the scan name; click on the *Open* command button.

To optimize a scan:

- From the *OPTIMIZE* menu, select the *Origin* command; Clip scan at origin
- From the *OPTIMIZE* menu, select the *Front* command; Clip the scan at front
- In the DISPLAY window, press the right mouse button, select the *Zoom* command
- Box the area of interest
- From the *OPTIMIZE* menu, select the *Snip at View* command
- From the *OPTIMIZE* menu, select the *Sub Horizontal Baseline*, *Construct Baseline*, or *Subtract Background* command

To smooth a scan:

- From the ENHANCE menu, select Criteria
- In the Set Criteria dialog box, enter Window Size, and Iterations Value
- From the ENHANCE menu, select the Smooth command

To Identify and integrate peaks:

- From the PEAKS menu, select the Peak Criteria command
- In the Peak Criteria dialog box, enter minimum Amplitude, Width and Area.
- From the PEAKS menu, select Auto Identify

- Position cursor on peak of interest
- From the PEAKS menu, select the Peak Info command
- From the PEAKS menu, select the Integrate Interactive command
- Drag the cursor from origin to front of the peak of interest

To edit peaks:

- From the PEAKS menu, select the Edit. . . command
- From the EEDIT menu bar select ADJUST, CREATE, OR DELETE

To save:

- From the SCAN menu, select the Save command

To generate a standard curve:

- From the MODE menu bar, select the Weight command
- If your standard is not a Bio-rad standard, from the FUNCTIONS menu, select Protein Editor; enter the names and molecular weights for peaks on the standard scan
- From the FUNCTIONS menu, select Calculate Standard; in the dialog box, select the file holding the table of names and weights.
- In the dialog box, select the scan of the standard proteins; associate names/molecular weights with peaks.
- In the dialog box, click on the Calculate command button; check regression type; click on the Curve command button; click on the Done command button.

To calculate the molecular weight of an unknown:

- From the MODE menu, select the Weight command.
- From the FUNCTIONS menu, select the Calculate Unknown command: open the file in which the unknown and standard protein scans are stored; select standard protein scan for which curves have been generated, select unknown protein scan, click on the Calculate command button

To print:

- From the MODE menu, select the Report command
- From the FUNCTIONS menu select the Contents command; select the type of report desired; set parameters
- From the FUNCTIONS menu, select the Print command

To transfer data from densitometer to the 1-D program:

Prepare the video densitometer:

- Choose DIRECTORY, choose #19, set RS-232 and Register =Current
- Choose DIRECTORY, CHOOSE #18, set Baud rate and even parity
- Choose DIRECTORY, choose #12, set filter frequency to none, Enhancement frequency to none and Boost factor at 0
- Choose Directory, choose #13, choose baseline subtract off
- Choose Directory, choose #1, choose #4 Calibrate

Prepare the 1-D program for data capture:

- From the MODE menu, select Capture
- In the dialog box, click on the Create command button, name the file in which the captured scan will be stored (include .SCN extension)
- Click on the Root Name edit field, name and identify the scan to be captured
- Click on the Save Parameters command button.
- In the transfer field, select appropriate Communications Port, Baud Rate and Parity

Capture data:

- Choose #1 Take Data/Operations, from the densitometers take data menu
- Note messages in Capture status box

To delete scans:

- From the DISPLAY menu, select the Delete command

Section 3

1-D Analyst II Software Functions

In this section, each of the 1-D Analyst II software functions and their command descriptions will be described in detail. All the 1-D Analyst II software functions can be accessed from the Mode Menu in the Main menu screen and throughout the program.

3.1 CAPTURE Function

The CAPTURE function of the software collects, tags, and stores one-dimensional data from the Bio-Rad Model 620 Densitometer. The CAPTURE function automatically saves the captured scans to the hard disk, making them available for display and subsequent analysis. To successfully capture data from the Bio-Rad Model 620 Densitometer, (OUTPUT DEFINITION) in the densitometer's DIRECTORY MENU must be set to "RS-232 Current", and (RS-232 COMMUNICATIONS) in the densitometer's DIRECTORY MENU must be set to "19,200 baud" and "even parity". When the CAPTURE function is selected the following screen appears on the PC monitor:

Capture Scan (Untitled)

Scan
Root Name:
Type:
Background ☐
Standard ☐
Unknowns ☒

Densitometer Settings
Filter Freq:
Enhancement Freq:
Boost Factor:
Baseline Subtracted ☐
Min: Max:
X(MM): Y(OD):
Track Loading:
Save Parameters

File
Create Append
Interpolation ☒ On

Transfer
Set Comm Capture Stop
Exit Cancel

Notes

Status
Select a File for Saving Scan Data

Command Descriptions

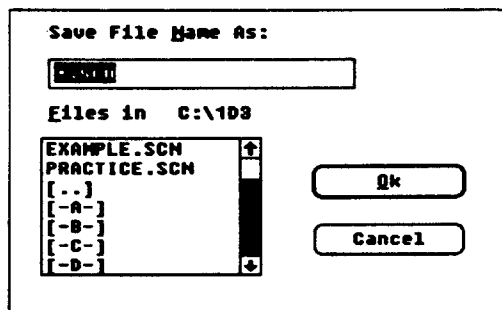
CAPTURE SCAN Screen

The commands within the capture screen identify the communications port and speed at which the densitometer will transfer the data, the scanfile name used to store the received scans, information about the scan type, and the densitometer settings used when the scan was digitized.

FILE Commands

1. FILE Command: *Create*

This command button allows you to name the scanfile which will contain the scans to be received from the Model 620 Densitometer. When *Create* is selected, the following dialog box appears on the screen:



This dialog box is a standard Microsoft Windows entry box. It provides a scrolling list box of the scanfile names within the current drive and directory and an entry box for entering the scanfile name to save the captured scans. Notice that the entry box will have **"*.SCN"** highlighted, indicating that the .SCN extension must be used to identify scanfiles. Use the keyboard to type an identifying scanfile name with the extension **".SCN"** [ENTER], e.g., **DATAGEL1.SCN** [ENTER].

Changing destination of captured scans

To change the destination of the captured scans from the default drive and directory (the drive and directory where the 1-D Analyst II program is located) to another drive or directory, perform one of the following:

Changing directories within current drive:

To change directories within the current drive, using the scroll bars if necessary, double click on **[. .]**. The list box will now list the directories available on the current drive. Double clicking with the left mouse button, on the desired directory will open that directory and list the scanfiles within that directory which have the scan scanfile extension (**.SCN**). Enter the scanfile name to save the captured scans as described above.

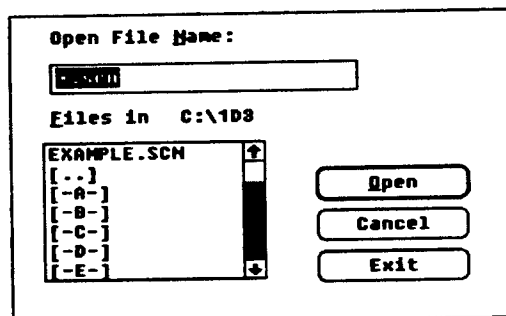
Changing current drive:

To change from the current drive (i.e., **[-C-]** drive), to another internal drive (i.e., **[-D-]** drive), or an external drive (i.e., **[-A-]** or **[-B-]** drives), use the scroll bars if necessary to scroll the list box window until the drive letters are visible. Now double click on the desired drive. The list box will now update to reflect the directories and scan scanfiles within that drive. Change directories and/or enter the scanfile name to save the captured scans as described above.

Once a scanfile in which to save the captured scans has been named, the Status box will prompt the user to: **"Enter and Save the Scan Parameters,"** and items within the Scan Parameter box will turn active allowing data to be entered.

2. FILE Command: Append

This command button allows you to append an existing scanfile which already contains scans previously captured with the 1-D Analyst II program. When *Append* is selected, the following dialog box appears on the screen:



This dialog box is a standard Microsoft Windows entry box. It provides a scrolling list box of the scanfiles names within the current drive and directory, and an entry box for entering the scanfile name to open and append the captured scans to. Notice that the entry box will have "*.SCN" highlighted, indicating that the .SCN extension must be used to identify scanfiles. Use the keyboard to type an identifying scanfile name with the extension ".SCN" [ENTER], e.g., DATAGEL1.SCN [ENTER], or move the mouse pointer to the scanfile name within the list box that you wish append, and double click with the left mouse button.

To append scanfile residing in another directory or drive follow the procedure outlined under the Create command.

After opening a scanfile to append, the Status box will prompt the user to: "Enter and Save the Scan Parameters," and items within the Scan Parameter box will turn active allowing data to be entered.

INTERPOLATION Command:

Scans exported by the Model 620 Densitometer to the 1-D Analyst II program contain 3,320 data points, half of which (every other point) is an interpolation provided by software on the densitometer. The interpolated points assist in visualization and automatic identification of peaks by removing the "stairstep" effect often seen.

With the interpolation command, the user can "turn off" the interpolation, in other words, the 1-D Analyst II software will remove every other data point leaving only the raw data points (1660) acquired by the Model 620 Densitometer's detector. This feature is offered for users requiring completely unprocessed raw data. It is however recommended that the interpolation be used as this feature facilitates the data analysis process.

TRANSFER Commands

1. TRANSFER Command: *Set Comm*

This command specifies the communications parameters for serial transmission of data from the densitometer to the PC. Generally, the densitometer interface cable (note: this is not a standard RS-232 interface cable, please consult the Cable Pin Out table in Appendix B) is connected to serial communications port number one (COM1) and other peripherals such as the serial mouse are connected to serial communications port number two (COM2).

Changing the communications parameters

The default communications parameters are:

Port: Com1

Baud Rate: 19,200

Parity: Even

To change any of these parameters, use the mouse to point and click on the appropriate radio button. When finished, point and click on the *OK Command Button*.

2. TRANSFER Command: *Capture*

This command prepares the software to receive a scan from the Bio-Rad densitometer. The *CAPTURE* command button will remain inactive (grayed) until a scanfile is created or appended, and the scan parameters are entered and saved. Once the scan parameters have been entered and saved, the *CAPTURE* button can be pressed. After the *CAPTURE* button has been pressed, the Status box will prompt the user to: "Output Data from the Densitometer". If the *CAPTURE* button is pressed before the scan parameters have been saved the following message will appear in the Status box: "Save the Scan Parameters Before Capturing Data."

3. TRANSFER Command: *Stop*

This command is only active after the *Capture* button has been pressed. The *Stop* command is used to interrupt the data capture process and exit from the data capture mode.

EXIT Command:

This command allows the user to exit from the *CAPTURE* window at any point except when in the capture mode.

CANCEL Command:

This command allows the user to cancel any operation within the Capture window which has already been initiated, and immediately exit the *CAPTURE* window.

Scan Parameters Field

For the Scan Parameters Field to be active, a Scanfile must first be created or appended by using either the *CREATE* or *APPEND* commands in the *FILE* command box.

1. Scan Parameters: *Root Name Entry Field*

The *Root Name Entry Field* allows up to fifteen characters to be entered which will be used to tag each individual scan as it is captured from the densitometer. Once the **SAVE PARAMETERS** command is pressed, all subsequently captured scans will have the same root name until the root name is changed and the scan parameters are saved again. Scans captured under the same root name are logged sequentially with a three digit extension starting with ROOTNAME.000. Once a scan is captured with a given root name, that root name and numerical extension will never change. If the scan is processed in any manner, an alphabetical appendix is attached to the root name and extension. For example, if ROOTNAME.000 were smoothed and then saved, the letter "a" would be appended to the root name and extension: i.e. ROOTNAME.000-a. The same scan can be save up to 26 times after an operation has been performed upon it.

2. Scan Parameters: *Type*

The Scan Type radio buttons allow each scan to be identified as a Background, Standard, or Unknown scan by simply clicking on the corresponding radio button. Designation of the particular scan type is important; the 1-D Analyst II software will use this information to route scans through its filing system. For example, a background scan must be identified as such in order for the program to use it in the SUBTRACT BACKGROUND function. The following is description of the three different scan types:

Background- A Background scan is data from an area of the one-dimensional gel, film, etc., which does not contain sample (unknown) only background data. Background data is scanned and exported to the 1-D Analyst II software in the same manner as the unknown data, therefore in order for the 1-D Analyst II software to differentiate it as a background scan it must be "tagged" as such.

Standard- A Standard scan is data from a one-dimensional SDS-PAGE gel which contains purified proteins of known molecular weight, such as the Bio-Rad Low Molecular Weight standards. Once the Standard scan data is captured by the software and "tagged" as a Standard scan, it can then be used by the software for the generation of a standard curve. This Standard curve is then used in the determination of molecular weights for unknown samples. The Standard scan type identification should only be used with SDS-PAGE protein standards data as the standard curve generated is log-linear.

Unknowns- The Unknown scan is any one-dimensional data which contains sample data, and is not a standard scan. If the Unknown scan contains protein data from an SDS-PAGE gel, it can be compared against a molecular weight standard curve, generated from standard data within the same gel, to determine the molecular weights of unknown proteins.

3. Scan Parameters: *Densitometer Settings*

The Model 620 densitometer has on-board firmware which can be used to manipulate and analyze the scan information before it is exported to the 1-D Analyst II software. The *Densitometer Settings* entry fields and check boxes allow the input of densitometer settings associated with each scan. Documentation of the operations

performed on each scan prior to export to the 1-D Analyst II software is essential as they will effect the interpretation of the scan data. **Note:** it is recommended that only the "raw" unprocessed scan be exported to the 1-D Analyst II software. The 1-D Analyst II software offers the same processing capabilities as the densitometer. And the raw scan data will be permanently archived on the computer hard disk.

Filter Frequency - If you have used the filtering option to smooth your data before exporting it to the 1-D Analyst II software you must enter the exact filter value used (0.05, 0.1, 0.2, 0.5, 1.0, 2.0, 4.0) for proper scan documentation.

Enhancement Frequency - If you have used the enhancement option to digitally enhance your data before exporting to the 1-D Analyst II software, you must enter the value used (0.05, 0.1, 0.2, 0.5, 1.0, 2.0) for proper scan documentation.

Boost Factor - If you have used the Boost option to control the amount of enhancement of high frequency components on your data before exporting to the 1-D Analyst II software, you must enter the value used (0-10) for proper scan documentation.

Baseline Subtracted - Click on the *Baseline Subtracted* check box if you have subtracted a baseline or background data using the densitometer's baseline subtraction feature. It is recommended that background data be taken using the densitometer's TAKE DATA/OPERATIONS function and exported to the 1-D Analyst II software (identified as a background scan by clicking on "Background" under *Scan Type*). This scan can then be subtracted from a standard or unknown scan using the *Sub Background* command.

Axis Minimums and Maximums - If the scan being exported from the densitometer has been zoomed in, using the ZOOM function on the densitometer's Operations Screen, the X and Y axis minimums and maximums should be entered in the the appropriate box. **Note:** it is recommended that the scan be zoomed out before it is exported so that the original unprocessed scan can be permanently archived on the external computer hard disk.

If the densitometer's field of view has changed from 200 mm to 80 mm by use of a High Resolution Lens, then we must convey this information to the 1-D Analyst II software by entering 0 in the X axis minimum box, and 80 in the X axis maximum box.

4. TRACK LOADING Entry Box:

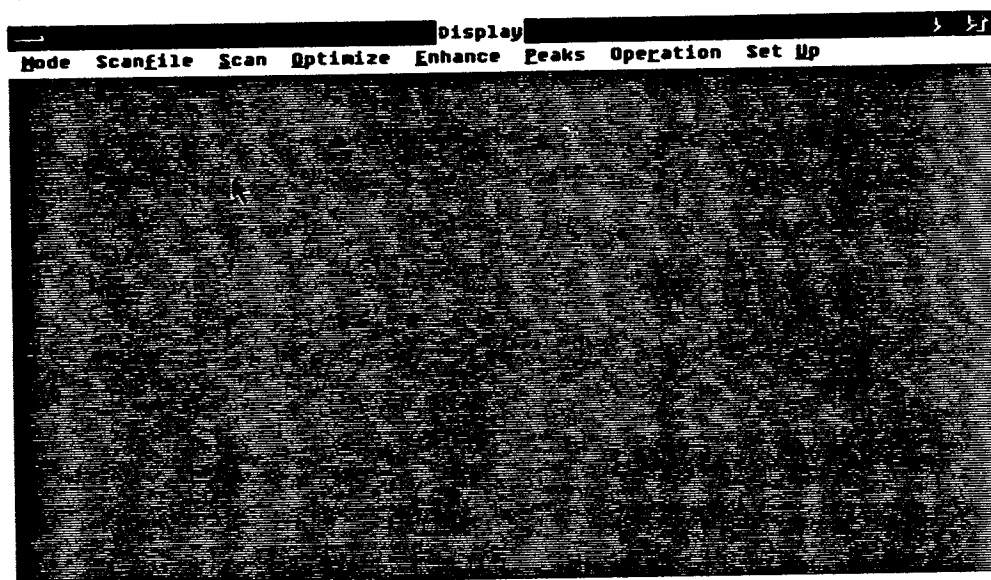
If the total amount of material loaded on a lane of the electrophoretic gel is known, the 1-D Analyst II software will calculate the amount of material within each identified peak based on its percentage of the total area under the scan curve. For example, if a 100 micrograms of protein were loaded on a gel lane before electrophoresis, and an individual peak within the scan contained 25% of the total scan area, then the software would report that peak to contain 25 micrograms of material. To enter a track load for a given scan, type in the amount of material loaded along with appropriate concentration units.

5. **SAVE PARAMETERS** Command:

After all the above mentioned scan parameter entries and selections have been made, the **SAVE PARAMETERS** Command button must be clicked on twice before data can be captured. After the first scan is captured, each subsequent scan will be captured with the same scan parameters. To change any scan parameters prior to capturing a scan, simply make the appropriate changes and click on the **SAVE PARAMETERS** Command button twice.

3.2 **DISPLAY Function**

The **DISPLAY** function of the software contains the graphic interface which is used to display and analyze stored scan data. Within the **DISPLAY** function are analytical tools for optimizing and enhancing the scan data, as well as advanced algorithms for the identification of peaks. When the **DISPLAY** function is selected the following screen will appear on the PC monitor:



Display Function Terminology

"Active Scan" - The only scan which is currently "live" and can have any function or operation performed upon it. On a color monitor, the active scan is denoted by its color (magenta unless the default colors have been changed). On a monochrome monitor, the active scan is denoted by its shade which can be determined by clicking on *Atv Scan* under color set up. To make a scan active when multiple scans are displayed, simply double click on the desired scan.

"Active Graph" - When more than one graph is displayed on the screen, the active graph will be the graph containing the active scan. To make a different graph active, simply double click on one of the scans within that graph.

"View Modification" - A view modification is a modification which affects only the display or viewing of the data, but does not actually modify the data itself. Any subsequent operation will effect the whole scan, not just the section currently in view.

"Data Modification" - A data modification is a modification which not only affects the viewing of the data, but actually transforms the data itself. All operations performed after the data modification will affect only the portion of the scan currently in view.

Command Descriptions

MODE Menu

The mode menu is continuous throughout the software program, allowing the user to move rapidly from one function to another without having to go through multiple menu levels. Also available in the *MODE* Menu is the *Exit* command, which allows the user to exit to the main menu instead of to another function.

SCANEILE Menu

These commands allow the deletion of scans within files and subsequent compacting of files to free up disk space. The program does not automatically compact files immediately after scan deletion as this can take significant processing time away from more important operations. Instead, the program will wait until the 1-D program is quit to automatically compact files which have had scans deleted from them. If disk space is required before quitting the program, then the *Compact Scanfile* command can be used.

1. SCANEILE COMMAND: Delete Scan . . .

When this command is selected, the Windows entry box will appear which provides; a scrolling list box of the scanfile names within the current drive and directory, and an entry box for entering the scanfile name to open for deleting scans. Notice that the entry box will have **"*.SCN"** highlighted, indicating that .SCN must follow a scanfile name. Use the keyboard to type an identifying filename with the extension ".SCN" [ENTER], e.g., *DATAGEL1.SCN* [ENTER], or move the mouse pointer to the scanfile name in the list box containing the scan(s) that you wish to delete, and double click with the left mouse button.

Locating Scanfiles in other drives and directories

Changing directories within current drive:

To change directories within the current drive, using the scroll bars if necessary, double click on [..]. The list box will now list the directories available on the current drive. Double clicking on the desired directory will open that directory and list the existing scanfiles within that directory which have the scanfile extension (.SCN).

Changing current drive:

To change from the current drive (i.e., [-C-] drive), to another internal drive (i.e., [-D-] drive), or an external drive (i.e., [-A-] or [-B-] drives), use the scroll bars if necessary to scroll the list box window until the drive letters are visible. Now double click on the desired drive. The list box will update to reflect the directories and scanfiles within that drive. Change directories and/or double click on the scanfile name, to open and delete scans.

After opening a scanfile for deletion of scan(s), a new dialog box will appear listing all the scans within the scanfile. To delete a single scan, point and click once on the scan name to highlight it, then click on the *DELETE* Command Button. To delete more than one scan at a time, press and hold [SHIFT] while pointing and clicking on all the scans

you wish to delete. After deleting the scan(s), the entry box will reappear again listing all the scanfiles within the current directory. If you wish to delete additional scans from another scanfile, point and click on the appropriate scanfile and repeat the same procedure. When you have finished deleting scans, point and click on the *CANCEL* or *EXIT* Command Buttons. It is important to note that at this point the scan names have been deleted from the scanfile, however, disk space occupied by the deleted scans will not be freed until the *Compact Scanfile* command is executed as outlined below.

2. SCANFILE COMMAND: *Compact Scanfile*

Disk space occupied by scans deleted using the *Delete Scan . . .* command is not freed up until either the 1-D Analyst II program is exited, or the *Compact Scanfile* command is used. By using the *Compact Scanfile* command, it is possible to free up the disk space occupied by the deleted scans without exiting the 1-D Analyst II software program. To compact scanfiles, simply point and click on *Compact Scanfile* in the *SCANFILE* menu, and the scanfiles will be immediately compacted.

SCAN MENU

These commands specify the movement of scans in and out of the graphic interface.

1. *SCAN* COMMAND: *New . . .*

The *New . . .* command will display a new graph containing a selected scan. A maximum of four different graphs can be displayed at once, with each graph containing only one scan.

When this command is selected, the Windows entry box will appear which provides a scrolling list box of the scanfile names within the current drive and directory, and an entry box for entering the scanfile name to open for displaying scans. Notice that the entry box will have *".SCN"* highlighted, indicating that *.SCN* must follow a scanfile name. Use the keyboard to type an identifying filename with the extension *".SCN"* [ENTER], e.g., *DATAGEL1.SCN* [ENTER], or move the mouse pointer to the scanfile name in the list box containing the scan that you wish open for display, and double click with the left mouse button.

Locating Scanfiles in other drives and directories

Changing directories within current drive:

To change directories within the current drive, using the scroll bars if necessary double click on *[.]*. The list box will now list the directories available on the current drive. Double clicking on the desired directory will open that directory and list the existing scanfiles within that directory which have the scanfile extension (*.SCN*).

Changing current drive:

To change from the current drive (i.e., *[-C-]* drive), to another internal drive (i.e., *[-D-]* drive), or an external drive (i.e., *[-A-]* or *[-B-]* drives), use the scroll bars if necessary to scroll the list box window until the drive letters are visible. Now double click on the desired drive. The list box will update to reflect the directories and scanfiles within that drive. Change directories and/or double click on the scanfile name, to open for display.

After you have selected a scanfile, the software will display the following scan-information dialog box:

Scan Name:

Scans in EXAMPLE.SCM

- ☒ background.000
- ☐ BRstandards.000
- ☐ sample.000

Type

- ☒ Raw
- ☐ Processed
- ☐ Standard
- ☒ Background
- ☐ Unknowns

Operations

- ☐ Subtracted From
- ☐ Enhanced
- ☐ Clipped
- ☐ Peaks Id'd
- ☐ MW Calculated

The scan-information dialog box contains a list box of all of the scans contained within the scanfile which has been selected. After you point and click on one of the scan names within the list box, the Type and Operations boxes will now show all the identifying characteristics of the scan. The Type box provides the following information:

Type Box

Raw - This identifies the highlighted scan as being a raw, unprocessed scan. This type of scan is always the original scan, as received from the densitometer. The 1-D Analyst II software treats the raw scan as sacred, saving all processed versions of the raw scan under an appended name. Raw scans will never have a character appendix.

Processed - This identifies the highlighted scan as being processed in some manner, differentiating it from its original raw form. Processed scans will always have at least one of the operations in the operations box highlighted, identifying the manner in which it has been processed. Processed scan names will always be followed by a character appendix, identifying it as being a processed scan. Appendices indicate the order in which processed scans were saved. For example, *CONTROL.001-a* would be the first processed version of the raw scan *CONTROL.001* to be saved.

Standard - This identifies the highlighted scan as having been tagged as a standard scan when it was captured from the densitometer. The processed standard scan is used in the generation of a standard curve for molecular weight determination of unknown samples. The Standard scan identification is only used for SDS-PAGE protein standards data as the standard curve generated is log-linear.

Background - This identifies the highlighted scan as having been tagged as a background scan when it was captured from the densitometer. The background scan is data associated with the background of the electrophoretic sample, but not the sample itself. A background scan can be subtracted from a standard or unknown scan to remove all area which is background data. A background scan can be displayed, but, the 1-D Analyst II software will not allow it to be processed in any manner.

Unknowns - This identifies the highlighted scan as having been tagged as an unknown scan when it was captured from the densitometer. An Unknown scan

is any one-dimensional data which contains sample data, and is not a standard scan. If the Unknown scan contains protein data from an SDS-PAGE gel, it can be compared against a molecular weight standard curve, generated from standard data within the same gel, to determine the molecular weights of unknown proteins.

Operations Box

Subtracted From - This identifies the highlighted scan as having had background area subtract from it. Three different types of background subtraction are available in the 1-D Analyst II software: 1) subtraction of a background scan, 2) subtraction of a constructed baseline, 3) subtraction of a horizontal baseline.

Enhanced - This identifies the highlighted scan as having been smoothed and/or enhanced. Smoothing is a weighted moving average which filters the high frequency noise associated with artifacts, such as scratches in film, etc. Enhance will increase the spatial frequency, assisting in the separation of partially resolved peaks.

To display a new graph containing a scan from within the scan list box, simply double click on the desired scan name, or enter the scan name complete with extensions and appendixes and then click on the **ADD** Command Button.

2. SCAN COMMAND: **Add** . . .

The **Add** . . . command will add a scan to an active graph (a graph containing an active scan as denoted by its color or shade). This command can be used until a maximum of four scans are displayed, in one or more graphs.

To add a scan to an existing graph, click on the SCAN menu then click on **Add** . . . Select a scanfile, then a scan for display as outlined above under **New**

3. SCAN COMMAND: **Delete**

The **Delete** command will delete only the active scan (denoted by its color or shade). To delete the active scan, simply click on the SCAN menu then click on **Delete**. The active scan will disappear from the graph window, and the scan which was added previously to the active scan will now become the active scan. If only one scan is displayed then deleted, the graph will close.

4. SCAN COMMAND: **Close Graph**

When this command is selected, the active graph, the graph containing the active scan, will close with all its contents. To close a graph which does not contain an active scan, double click on one of the scans within that graph making it active and then select the **Close Graph** command.

5. SCAN COMMAND: **Original**

When this command is selected, the program will remove the active scan and replace it with the original version of the scan. The **Original** command is necessary to restore a scan to original form once any "data modification" has been performed upon it. For example, using the **Zoom** function to get a closer view of a section of a scan is a "view modification" which can be restored by pressing the right mouse button and clicking on

Restore, which will restore the graph to the previous unzoomed view. If however, the *Snip at View* command was used after zooming in on a section of the scan, a "data modification" of the scan has been performed. The *Original* command would now be required to restore the scan to its original form.

6. SCAN COMMAND: *Previous*

When this command is selected, the program will toggle back and forth between the current and the previous scans which has had either a data or view modification performed upon it.

7. SCAN COMMAND: *Save*

When this command is selected, the active scan will be saved with any modifications which have been performed. Using the *Save* command will not write over the original scan data, instead it will save the processed scan using the same scan name along with a character appendix, i.e., *CONTROL.001-a*. Note: The *Save* command is executed quickly. No visible notice that the operation has been successfully completed, such as an hour glass, is provided. If you are not sure that the scan has been saved, simply click on *Save* again. The program will ask you if you wish to overwrite existing scan. Click on the *Cancel* command button to proceed.

8. SCAN COMMAND: *Save As . . .*

When this command is selected, the active scan can be saved using any root name desired up to fifteen characters in length. The program will automatically add the appendix "a" identifying the scan as a processed scan. Note: no extension is necessary when enter the scan name, as the program will automatically add the following extension and appendix ".000-a".

9. SCAN COMMAND: *Scan Info*

When this command is selected, a "hand pointer" icon will appear for pointing and clicking on any scan within the active graph. By placing the "finger" of the icon on one of the scans in the graph, and clicking with the left mouse button, a table of information regarding that scan will appear on screen. Pointing and clicking with the icon on another scan within the active graph will update the scan information table to reflect that scan. This command is not only useful for identifying the name of a scan within a graph, but also for quickly providing vital information about each scan. The scan information table can be moved about the screen, to uncover scan portions, by clicking on the title bar and dragging the box to a new location. When finished with the *Scan Info* command, click on the *OPERATION* menu, then click on either *Complete* or *Cancel* to remove the scan information box.

OPTIMIZE MENU

These commands provide various scan optimization routines enabling the user to extract the most information possible from scan data. Commands within the *OPTIMIZE* menu act upon the scan(s) as an entirety, not upon individual data points within the scan.

1. OPTIMIZE COMMAND: Snip at View

When this command is selected, any view modifications performed with the Zoom function will be transformed into a data modification. The user can therefore zoom in on any section of the displayed scan and analyze that section of the scan exclusively by pointing and clicking on the Snip at View command. Once the Snip at View command has been used, either the Original or Previous command must be used to restore the original data or toggle back to the last data modification.

2. OPTIMIZE COMMAND: Construct Baseline

When this command is selected, the user is allowed to construct an interactive baseline to accommodate scan background which is not perfectly horizontal. This command is frequently used for constructing a "valley-to-valley" baseline whereby the scan background envelope is removed by connecting the valley of a peak to the valley of an adjacent peak.

To use the Construct Baseline command, simply move the arrow icon to a starting horizontal (X Minimum) position, which is outside the left edge of the scan (origin), and click the left mouse button. A dot will identify this first baseline location. Continue constructing the baseline by clicking with the left mouse button at each desired location, progressing towards the right end of the graph (X Maximum). The program will connect the points as they are constructed. The beginning and ending points of the constructed baseline must be outside the constraints of the scan. If the baseline constructed is not correct, simply click on Cancel under the OPERATIONS menu to remove the baseline, and repeat the steps outlined above to construct a new baseline. To subtract the constructed baseline, click on Complete under the OPERATIONS menu, or click on Complete under the OPERATIONS menu and then on Sub Constructed Baseline under the OPTIMIZE menu, depending on whether the Subtract Baseline is selected from the GRAPH SET UP menu. The resulting graph displayed will be only that area above the constructed baseline, as all area below the baseline will be subtracted.

3. OPTIMIZE COMMAND: Sub Construct Baseline

The Sub Construct Baseline command is only operational if the Subtract Baseline Check Box is deactivated under the SET UP Menu, and if a baseline has been constructed using the Construct Baseline command. By deactivating the Subtract Baseline Check Box, the subtraction of the constructed baseline is now a three step process versus the default two step process. The constructed baseline will then be filled in and remain visible after the Complete command, under the OPERATION Menu has been selected. This allows the constructed baseline to be displayed in the same manner as a scan for comparison. To perform the subtraction of the constructed baseline from the active scan, simply point and click on the Sub Construct Baseline command.

4. OPTIMIZE COMMAND: Sub Horizontal Baseline

When this command is selected, a horizontal baseline can be selected and automatically subtracted from the active scan. This command is useful for subtracting background from scans such as autoradiograms containing a uniform background level.

To subtract a horizontal baseline from the active scan, simply point and click on *Sub Horizontal Baseline*, the mouse pointer will change to a horizontal line which is adjustable in the vertical direction. Move the horizontal line cursor up or down until the desired baseline position is located, then click on the left mouse button to automatically subtract the horizontal baseline.

5. OPTIMIZE COMMAND: *Subtract Background*

When this command is selected, actual background data captured from the densitometer can be subtracted from the active scan. This method of background subtraction is usually the most accurate and should be the method of choice when available.

To perform the subtraction of a background scan from an active scan, click on the OPTIMIZE Menu, then click on the *Subtract Background* command. Select the scan file containing the desired background scan(s) as outlined above under the SCAN Command. Only scans designated as background scans at the time of capture from the densitometer will be available for subtraction. Once a background scan is selected it is immediately subtracted from the active scan, and the resulting scan is then displayed.

6. OPTIMIZE COMMAND: *Difference Scan*

The *Difference Scan* command allows the subtraction of any scan (except a background scan) from the active scan, creating a difference Scan. This Difference Scan can subsequently analyzed in the same manner as an original scan. The *Difference Scan* command is useful for the determination of the difference between two scans such as a control and an experimental sample.

To subtract one scan from another both scans must be displayed in the same graph. When two scan are displayed in the same graph, the program will automatically subtract the non-active scan (subtrahend) from the active scan (minuend). If more than two scan are displayed in the same graph, when the *Difference Scan* command is selected a hand will appear. Point the hand to the scan which you wish to use as the subtrahend and click the left mouse button. Once the difference scan has been executed, the resulting difference scan will be displayed as the active scan.

7. OPTIMIZE COMMAND: *Origin*

The *Origin* command allows the identification of the gel lane origin (point of sample application/beginning of resolving gel) within the scan. The *Origin* command is used in conjunction with the *Front* command for the determination of peak relative mobility, and subsequent molecular weight analysis for SDS-PAGE gels.

To specify the origin of the displayed active scan, click on the OPTIMIZE Menu, then click on *Origin*. The mouse pointer will change to a scissor icon which should be placed near the left edge of the scan (X min.), on or above the origin. Press the left mouse button and the portion of the scan left of the origin will be deleted with the point of "cut" now being the new origin. Note: if the scan was not originally scanned in the correct direction, i.e., the origin is near the right edge (X max.) of the scan, then the Flip Command described below should be used before the origin and front are identified.

8. OPTIMIZE COMMAND: *Eront*

The *Eront* command allows the identification of the gel lane front (location of the non-restricted dye front) within the scan. The *Eront* command is used in conjunction with the *Origin* command for the determination of peak relative mobility, and subsequent molecular weight analysis for SDS-PAGE gels.

To specify the front of the displayed active scan, click on the OPTIMIZE Menu, then click on *Eront*. The mouse pointer will change to a scissor icon which should be placed near the right edge of the scan (X max.) on or above the front location. Press the left mouse button and the portion of the scan right of the front will be deleted with the point of "cut" now being the scan front. Note: if the scan was not originally scanned in the correct direction, i.e., the origin is near the right edge (X max.) of the scan, then the *Flip* Command described below should be used before the origin and front are identified.

9. OPTIMIZE COMMAND: *Flip*

The *Flip* command is used to flip a scan which has been scanned in the incorrect orientation, with the front near the left edge of the scan. If an incorrectly orientated scan is not flipped, the calculated relative mobility values would be erroneous.

To flip a scan simply click on the OPTIMIZE Menu, then click on *Flip*. The data point sequence of the displayed active scan will then be reversed, with the origin of the scan now located near the X minimum location.

10. OPTIMIZE COMMAND: *Align 2 pts*

The *Align 2 pts* command allows two overlayed scans to be aligned along the X-axis in relation to two corresponding points on each scan. By aligning the overlayed scans, the differences in scanning position or gel shape within a gel, or among gels can be compensated. Thus comparison of peak mobilities among different scans is facilitated with the use *Align 2 pts* command.

To align any two scans which are overlayed in the same graph, click on the OPTIMIZE Menu, then click on *Align 2 pts*. The mouse pointer will be replaced by a downward pointing hand. Move the "finger" of the hand to a point on one of the scans, such as the top of a peak, which you wish to align with a point on the other scan, and click the left mouse button. A small circle will remain identifying the first point location. Now move the hand to a desired or corresponding point along the second scan, and click or press and release the left mouse button. Once a point on the second scan has been identified, the program will automatically align the two scan along the X-axis.

11. OPTIMIZE COMMAND: *Align 4 pts*

The *Align 4 pts* command allows two scans to be aligned not only along the X-axis, but additionally along the Y-axis so that the scans can be precisely overlayed on top of each other. By providing two "landmark" points along each scan, the *Align 4 pts* command uses a stretching routine, allowing for slight mobility differences between different gel lanes.

To align and stretch any two scans which are overlayed in the same graph, click on the OPTIMIZE Menu, then click on *Align 4 pts*. The mouse pointer will be replaced by a

downward pointing hand. Move the "finger" of the hand to a point on one of the scans, such as the top of a peak, which you wish to align with a point on the other scan, and click the left mouse button. A small circle will remain identifying the first point location. Now move the hand to a desired or corresponding point along the second scan, and click or press and release the left mouse button. Repeat this procedure with the remaining two points. Note that the order of identification of the four points on the scans is not important, i.e., two points can be identified on the first scan before any point is identified on the second scan. Once all four points have been identified, the program will automatically align the two scans stretching the second scan as necessary.

12. OPTIMIZE COMMAND: DragXY

The DragXY command allows an active overlaid scan to be dragged in any direction around the graph. This can be useful for organizing many scans within the graph to facilitate interpretation. For instance four scans can be placed one slightly above the other for simplified visualization of scan differences.

To drag an active scan in a XY direction, click on the OPTIMIZE Menu, then click on DragXY. The mouse cross hair will be replaced by a vertical fish hook icon. Move the fish hook icon to a location within the graph where you wish to move the active scan, and click the left mouse button. The active scan will now appear at the new location. By pressing the left mouse button instead of clicking, a connecting line will be drawn from the scan to the new location and the scan will not move until the mouse button is released.

13. OPTIMIZE COMMAND: DragX

The DragX command allows an active overlaid scan to be dragged only in the X (distance) direction along the graph. This can be useful for quickly aligning or organizing many scans within the graph to facilitate interpretation while maintaining the scans original Y (optical density) position. For instance, four scans can be overlaid on top of each other, then dragged along the X (distance) direction for simplified visualization of scan differences.

To drag an active scan in a X direction, click on the OPTIMIZE Menu, then click on DragX. The mouse cross hair will be replaced by a horizontal fish hook icon. Move the fish hook icon to a location within the graph where you wish to move the active scan, and click the left mouse button. The active scan will now appear at the new location. By pressing the left mouse button instead of clicking, a connecting line will be drawn from the scan to the new location and the scan will not move until the mouse button is released.

14. OPTIMIZE COMMAND: DragY

The DragY command allows an active overlaid scan to be dragged only in the Y (optical density) direction along the graph. This can be useful for quickly aligning or organizing many scans within the graph to facilitate interpretation while maintaining the scans original X (distance) position. For instance, four scans can be overlaid on top of each other, then dragged along the Y direction for simplified visualization of scan differences.

To drag an active scan in a Y direction, click on the OPTIMIZE Menu, then click on DragY. The mouse cross hair will be replaced by a vertical fish hook icon. Move the fish hook icon to a location within the graph where you wish to move the active scan, and click the left mouse button. The active scan will now appear at the new location. By pressing the left mouse button instead of clicking, a connecting line will be drawn from the scan to the new location and the scan will not move until the mouse button is released.

ENHANCE MENU

These commands provide two methods for enhancing low frequency electrophoretic data for improved data analysis. Both the Smooth and Enhance commands can bring insight to electrophoretic data by suppressing unwanted noise, and/or by boosting low frequency data.

1. ENHANCE COMMAND: Criteria

The Criteria command allows you to specify the level of smoothing or enhancement which you desire for your particular sample data.

Window Size- This entry field allows you to enter the filtering window size value used by the SMOOTH command. Enter a value from 3 (minimum smoothing), up to 99 (maximum smoothing). This value determines the number of data points which the moving weighted algorithm will use to smooth the data.

Iterations- This entry field allows you to enter the number of iterations (passes) you wish to have the SMOOTH command perform using the specified window size. Enter a value from 1 smoothing pass, up to a maximum of 99 smoothing passes. The higher the number of iterations, the more the scan will be smoothed.

En Boost- This entry field allows you to enter the enhancement boost factor. The boost factor is a constant which is multiplied by high spatial frequencies extracted from the data. The result is then added back into the original data. Enter a boost value between 1.0 (minimum) and 10.0 (maximum) as desired.

Display Each- When this check box is activated, the program will successively display each smoothed result scan when an iteration is specified. This feature enables you to observe the effect each smoothing pass has on the scan. This process of displaying the smoothing progression takes significantly more time than displaying the final result of all iterations, therefore if expediency is a factor this function should be deactivated.

To change any of the above values from their default setting (Window Size=3, Iterations=1, Enhancement Boost=1.0, and Display Each=deactivated), simply tab down through or point and click on each entry field and type in the desired value. When finished setting the desired parameters, click on the Set command button.

2. ENHANCE COMMAND: *Smooth*

When this command is selected, the program will smooth the active scan according to the window size and number of iterations assigned under *Criteria*. The algorithm used by the *Smooth* command is simply a moving weighted average of nearby data points. The window size (filtering frequency) determines the number of data points that will be involved. The number of iterations determines how many times the filter will be passed over the scan. The optimal window size and number of iterations must be empirically determined for your particular sample.

The smoothing feature is designed to suppress high spatial frequencies, which are usually associated with noise as opposed to the low frequency electrophoretic data. The larger the window size the more the data will be suppressed down to a lower frequency. The effect of suppressing higher frequencies is to yield a smoother trace. Normally, most electrophoretic data should be "lightly" smoothed (window size between 3 and 7) before further evaluation such as automatic identification of peaks. The application of a moving weighted average filter to smooth the original data will change the shape of the curve, however, it will not change the area under the curve. That is, for every peak reduced an equivalent amount of area is distributed to the surrounding valleys. Thus integrating the final smoothed scan will give the same result as integrating the original scan.

3. ENHANCE COMMAND: *Enhance*

When this command is selected, the program will enhance the active scan according to the enhancement boost factor assigned under *Criteria*. The enhancement feature is designed to boost high spatial frequencies, helping you identify and integrate only partially resolved peaks. This is done by extracting the high spatial frequencies, multiplying them by a constant (the boost factor) and adding them back into the original data. In the enhancement process, the initial filtering is performed which uses the enhancement frequency for its cutoff. The filtered result is then subtracted from the original, leaving what was originally suppressed by filtering, i.e., the high frequency components. Thus, we have extracted rather than suppressed components above the enhancement frequency.

Some notes on using the *Enhance* command:

Application of a weighted moving average filter, as used in the *Smooth* command, to the original data will not change the area under the curve. That is for every peak reduced, an equivalent amount of area is distributed to the surrounding valleys. Thus, integrating the final result will give the same result as integrating the original scan. Reasonable judgement must be exercised when setting the enhancement boost factor for the *Enhance* command. Specifically, if the enhancement yields a result at a given point which is above 3.0 O.D. or below zero, the value is clipped to whichever limit it transcends. This keeps values within the value range of the software, but does not affect the conservation of area as described above. The rule to keep in mind is not to use boost factors so high that the results go off scale. If this rule is followed, the measured areas of peaks will be accurate.

1. PEAKS COMMAND: Peak Criteria . . .

The Peak Criteria . . . command allows you to enter specific peak criteria parameters for use with the automatic peak identification routine. The criteria which you set are exclusion values which will determine which peaks within the scan will be identified and integrated. If no criteria are set (default values) the Auto Identify command will identify any signal within the scan it considers a peak regardless of its size. Any or all of the peak identification parameters can be used to eliminate unwanted peaks when using the automatic peak identification routine.

Amplitude- This entry field allows you to select a minimum peak amplitude (optical density), whereby all peaks with a maximum peak height below this value will not be identified. The amplitude parameter is the most expedient of the identification criteria as you can use the pointer to quickly determine a "cut off" optical density value for your peaks of interest, before entering this value in the Peak Criteria dialog box.

Width- This entry field allows you to select a minimum peak width (mm) "cut-off" value, whereby all peaks with a width less than this minimum value will not be identified.

Area- This entry field allows you to select a minimum peak area (O.D.*mm) "cut-off" value, whereby all peaks with an area less than this minimum value will not be identified.

Sm Win- The Smoothing Window entry field allows you to select a smoothing window value for use in the program's automatic peak identification routine. The smoothing window value is the number of data points within the scan (a full scan consists of 3320 data points), that the peak identification routine will use in the determination of a peak.

Using a smoothing window has the same effect as the Enhance Menu's Smooth command, as it eliminates unwanted peaks (noise) prior to peak identification. However, it does not change the actual data points as does the Smooth command. As an increasing smoothing window value is entered along with at least one iteration, fewer unwanted peaks (scan noise) will be identified as peaks. This allows you to adjust the automatic peak identification routine without having to do any or additional smoothing of the data points using the Smooth command. The optimal smoothing value must be empirically determined for your particular sample. Note: for the smoothing window to be operational a smoothing iteration value of at least one must be entered.

Sm Iter- The smoothing iterations entry field allows you to select the number of passes which the smoothing window will make over the scan. The more iterations that are performed, the more accurate the smoothing and subsequent peak identification will be. The computer's calculation time will, however, increase dramatically with each additional iteration that is performed. An iteration value of between 1 and 3 is usually sufficient. If the automatic peak identification routine is still identifying unwanted noise, either increase the smoothing window size, enter peak amplitude, width, or area criteria's, or perform an actual smoothing of the data points using the Smooth command.

Note: for the smoothing window to be in effect a smoothing iteration of at least 1 must be entered.

Calc'd Win- This message box displays the smoothing window value calculated by the program during the last automatic peak identification. This value will be zero when the program is first started, and can not be changed by the user as it only provides information regarding the smoothing window calculated during the last previous automatic peak identification. The calculated smoothing window value serves a guide to the user when setting the smoothing window value under *Sm Win*. If the automatic peak identification routine identified more or fewer peaks than meet your criteria, the calculated smoothing window value provides a guideline as to whether you should increase the smoothing window (identify less peaks) or decrease the smoothing window (identify more peaks).

To change any of the above values, except for *Calc'd Win*, from their default settings (0.00), simply tab down through or point and click on each entry field and type in the desired value. When finished setting the desired parameters, click on the *Set* command button.

2. PEAKS COMMAND: *Edit . . .*

The *Edit . . .* command allows you to manually create, adjust, or delete any peak within the active scan. When the *Edit . . .* command is selected, the PEAK EDIT window will replace the DISPLAY window, providing a new menu for peak editing. Note: it is not necessary to have previously identified any peaks using *Auto Identify*, as all peaks within the graph can be manually identified if desired. The following menu items are available within the PEAK EDIT window:

EDIT- The *EDIT* Menu contains the *Cancel* and *Exit* commands. Both commands will exit the PEAK EDIT window and return to the DISPLAY window. When the *Exit* command is used, all changes made in the PEAK EDIT window (creations, adjustments, and deletions) will be reflected in the DISPLAY window. Note that the peaks will not be renumbered until exiting the PEAK EDIT window. To cancel all editing done within the PEAK EDIT window and return to the DISPLAY window simply press the *Cancel* command.

CREATE- The *CREATE* menu/command allows you to quickly manually identify peaks within the active scan. When this menu/command is selected, the pointer will change to a "magic wand" icon. Simply move the point of the "magic wand" to the origin of the peak you wish to identify, and press the left mouse button. Continue pressing the left mouse button while dragging the "magic wand" to the right until the end (front) of the peak is reached. The area under the curve will fill in as the "magic wand" is dragged across the scan. If the boundaries of the peak you are creating overlap a previously created peak, the area of overlap will not be filled in. When finished creating peaks, either click on the *CANCEL* menu/command to use other editing commands, or click on the *Exit* command under the *EDIT* menu to return to the DISPLAY window.

ADJUST- The **ADJUST** menu/command allows you to quickly adjust the boundaries of a previously identified or created peak. When this menu/command is selected, the pointer will change to a "tuning fork" icon. Simply place the arrow head of the icon over the peak which you wish to adjust and press the left mouse button. The "tuning fork" icon will now change to a "magic wand" icon allowing you to redraw the area for that peak as outlined above under **CREATE**. When the mouse button is released after the adjustment, the icon will change back to the "tuning fork" allowing you to select another peak for adjustment. Continue this procedure until all peaks within the scan have been adjusted as desired. Click on either the **CANCEL** menu/command to use other editing commands, or click on the **Exit** command under the **EDIT** menu to return to the DISPLAY window.

DELETE- The **DELETE** menu/command allows you to rapidly delete any peak within the active scan. When this menu/command is selected, the pointer will change to a "hand grenade" icon. Simply place the arrow head of the icon over the peak which you wish to delete and press the left mouse button. The peak will immediately disappear as the graph is updated. Continue this procedure until all peaks within the scan have been deleted as desired. Click on either the **CANCEL** menu/command to use other editing commands, or click on the **Exit** command under the **EDIT** menu to return to the DISPLAY window.

CANCEL- The **CANCEL** menu/command allows you to cancel the current edit operation.

3. **PEAKS** COMMAND: *Auto Identify*

The *Auto Identify* command performs an automatic identification of all peaks within the active scan, which meet the criteria set under *Peak Criteria* . . . The automatic identification routine is designed to identify most if not all peaks within a variety of electrophoretic samples tested in our laboratories. For your particular sample it may be necessary to adjust the smoothing window size and number of iterations under *Peak Criteria* . . . , and/or manually edit the results. The integration algorithm used in the automatic peak identification program is a trapezoidal integration algorithm. To use the *Auto Identify* command, simply point and click on **PEAKS** and select *Auto Identify*. The length of time required by the computer to identify the peaks within the active scan will depend upon a variety of factors such as the speed of computer, complexity of the scan, and peak criteria set. The area values for each identified peak can be found using the *Peak Info* command or by saving the scan and printing a report.

4. **PEAKS** COMMAND: *Gaussian*

The *Gaussian* curve fitting command is used for the integration of overlapping peaks. Because overlapping peaks are not "baseline" resolved, using the integration values determined with *Auto Identify* command would not be precisely accurate. The *Gaussian* curve fitting command calculates what the normal progression of the peak would be if it were not smeared (completely resolved), providing for more accurate quantitation values. The *Gaussian* curve fitting command can only be used once the peaks of interest have been identified using the *Auto Identify*, or *Edit* . . . commands. Once all the desired peaks within the scan have been identified, simply point and click

on **PEAKS**, and select **Gaussian**. Instead of filling in the area under the curve, the **Gaussian** curve fitting command will overlay a different colored trace for the area boundary which was integrated for each peak. The area values for each identified peak can be found using the **Peak Info** command or by saving the scan and printing a report.

5. **PEAKS** COMMAND: *Integrate Interactive*

The *Integrate Interactive* command allows you to quickly get area values for any peak or segment of the active scan. These values are not retained anywhere within the program and must be written down if they are to be used further. To use the *Integrate Interactive* command, simply click on **PEAKS**, and select *Integrate Interactive*. The pointer will change to an "integral" icon. Place the icon at the beginning of the area you wish to integrate, and press the left mouse button. The area box in the lower right corner will now display an area value of zero. Drag the icon to the end of the area you wish to integrate and release the mouse button. The area box will now display the integral value for the area just identified. Every time the left mouse button is pressed the area value will zero, and a new area can be integrated by dragging the icon. Please note that the icon must be dragged from left to right for proper integration. To exit the *Integrate Interactive* command, click on **OPERATION** and select **Cancel**.

6. **PEAKS** COMMAND: *Peak Info*

The *Peak Info* command provides information regarding any peak which has been identified within the active scan. To use the *Peak Info* command, point and click on **PEAKS** and select *Peak Info*. The pointer will change to a "hand" icon. Using the icon, point to any peak area, and click in the left mouse button. The following dialog box will appear:

PEAK			
Number	5		
Name	Lysozyme		
Start	58.87	Mean	0.00
End	57.97	Std Dev	0.00
Amplitude	0.45	Amp	0.00
Amp X Loc	53.76		
Width	7.10		
Area	0.70		
Percent	19.05		
Distance	41.32		
RF	0.00		
MW	14400.00		
Amount			

The following are descriptions of the information provided in the Peak Information dialog box:

Number- This identifies which peak the peak information box is currently listing information for.

Name- If the peak is a standard protein from a standard scan which has had a molecular weight standard curve calculated, then name belonging to the peak would be displayed here.

Start- This identifies the x location (mm) where the peak starts (origin).

End- This identifies the x location (mm) where the peak ends (front).

Amplitude- This identifies the maximum peak height (O.D.).

Amp X Loc- This identifies the x location (mm) of the peak amplitude.

Width- This identifies the width of the peak (End minus Start).

Area- This identifies the area associated with the peak. In terms of absolute quantitation, this value would be equivalent to macromolecule concentration.

Percent- This identifies the percentage of the total area under the scan curve which is associated with that peak.

Distance- This identifies the distance (mm) the peak amplitude is from scan origin.

RF- This is the relative mobility value calculated for the peak. This is the ratio of the distances traveled by the spot and the dye front. This value is always less than one.

MW- This is the molecular weight value for the peak which was determined using the molecular weight function. A value will only be displayed if the molecular weight of the peak has been previously calculated.

Amount- The amount in peak will only be calculated and displayed here if a scan track loading amount was entered during the capture process. This amount in peak value is calculated from the percent total scan area attributed to this peak.

Mean- The mean value will only be displayed if gaussian curve fitting was performed. This is the mean x value (mm) for the gaussian curve.

Std.Dev- The standard deviation value is one standard deviation from the calculated mean. This value will only be displayed if gaussian curve fitting was performed.

Amp- The gaussian curve amplitude (max. curve height) will only be displayed here if gaussian curve fitting was performed.

1. OPERATIONS COMMAND: Complete

The Complete command will finish in any operation which is currently in progress. This command is used after manually constructing a baseline. Complete instructs the

program to fill in the constructed baseline or to go ahead and subtract the baseline from the active scan, depending upon whether the *Baseline Subtraction* box is checked under the *Graph Set Up* menu. Additionally, the *Complete* command will terminate the *Peak Info* and *Graph Info* commands.

2. OPERATIONS COMMAND: *Cancel*

The *Cancel* command will cancel any operation currently in progress. For example: if an error was made during the construction of a baseline, the command can be cancelled using the *Cancel* command so that a new constructed baseline can be started.

The *Cancel* command will be highlighted when available for cancelling a operation in progress. Simply click on the OPERATIONS Menu and select *Cancel* and the operation will be canceled, bringing the program back to its previous status.

1. SET UP COMMAND: *Colors*

The *Colors* command allows you to adjust the colors or greys (depending on whether you have a color or monochrome graphics hardware) of all scans and the graph background. This enables you to easily differentiate the scans. The default colors for the scans and graph background are: Scan 0= white, Scan 1= red, Scan 2= yellow, Scan 3= blue, Active scan= magenta, Graph background=black. A color range of 0 to 256 is provided for each Red, Green, and Blue color wheel. The shade values are permanently stored in the Microsoft Windows WIN.INI file so that shades need not be readjusted each time the 1-D Analyst II program used.

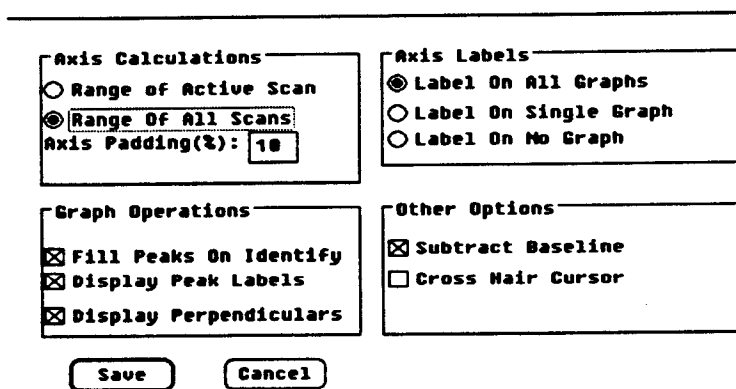
To change the color/grey shade, of the graph background or any scan, from its default value, perform the following steps:

- 1) Point and click on the *SET UP* Menu, and select the *Colors* command to display the Colors dialog box.
- 2) Click on the background or appropriate scan command button. The color window will now display the color currently assigned to that scan or background.
- 3) Adjust any or all of the 3 color wheels by pressing either the up or down arrow until the desired color/grey shade is displayed in shade window.
- 4) Set the color/grey shade by clicking on the *Fix* command button. Continue this procedure for any or all of the graph options as desired. When finished simply click on the *Exit* command button. The Colors dialog box will disappear and the current graph will update to reflect the color or grey shade changes.

To reset the color or grey shades to their default values, simply click on the *Default* command button, then the *Exit* command button. The Colors dialog box will disappear and the current graph will update to reflect the default colors.

2. SET UP COMMAND: Graph Setup . . .

The Graph Setup . . . command allows you to have complete control of the graph display parameters. By setting up the graph parameters exactly as you wish, you will be able to interpret your data faster and more easily. When this command is selected the following dialog box will appear:



The dialog box is titled "Graph Setup" and is divided into four main sections: "Axis Calculations", "Axis Labels", "Graph Operations", and "Other Options".

- Axis Calculations:** Contains three radio buttons: "Range of Active Scan", "Range Of All Scans" (which is selected), and "Axis Padding(%):" followed by a text box containing "10".
- Axis Labels:** Contains three radio buttons: "Label On All Graphs" (which is selected), "Label On Single Graph", and "Label On No Graph".
- Graph Operations:** Contains three checked checkboxes: "Fill Peaks On Identify", "Display Peak Labels", and "Display Perpendiculars".
- Other Options:** Contains two checkboxes: "Subtract Baseline" (checked) and "Cross Hair Cursor" (unchecked).

At the bottom of the dialog box are two buttons: "Save" and "Cancel".

Axis Calculations Field

Range of Active Scan- When this radio button is selected, the graph axis scaling will adjust according to only the active scan, and will not take into account the range of other scans within the same graph.

Range of All Scans- When this radio button is selected, the graph axis scaling will adjust according to the range of all scans within the graph, including the active scan. This is the default setting.

Axis Padding (%)- This entry field allows you to enter the percentage of the graph which you want to have as padding around the scans. If no padding is selected, the zero O.D. values for the scan(s) will lay directly on the X-axis. Some padding is recommended as it facilitates the use of graphic tools such as the Construct Baseline and Zoom tools. The default value is 10% padding.

Axis Labels Field

Labels On All Graphs- When this radio button is selected, axis labels will be drawn on all graphs when multiple graphs are displayed. This is the default setting.

Labels On Single Graph- When this radio button is selected, axis labels are drawn when a single graph is displayed, however, when a new graph is added no axes will be displayed on any of the graphs. This feature is useful as when four different graphs are displayed at the same time the screen becomes overwhelmed with information and the many axes can be distracting. By turning off the axes, extracting information can be easier.

Labels On No Graph- When this radio button is selected, no axis labels will be drawn on any of the graphs, even a single graph. This feature is useful for those that find the axes distracting.

Graph Operations

Fill Peaks On Identify- When this check box is checked, each peak on the graph will be filled in once it is identified using either the automatic or manual peak identification methods. If this box is not checked, peaks will not be filled in, and the only way to identify the peaks in the graph will be if the labels and perpendiculars are displayed. Note that when manually creating peaks using the Edit function the peaks will be filled in, however when exiting the edit function back to the display function the fill will disappear. The default setting is on.

Display Peak Labels- When this check box is checked the peak's labels, including peak numbers and molecular weight protein names, will be displayed on the graph above the corresponding peak. If this box is not checked, none of the peak labels will be displayed in the graph. The default setting is on.

Display Perpendiculars- When this check box is checked, perpendiculars will be displayed identifying the X (mm) boundaries for each peak. If this box is not checked, no perpendiculars will be displayed for the origin and front of each peak. The default setting is on.

Other Options

Subtract Baseline- When this check box is checked, the subtract constructed baseline function is changed from a three step operation to a two step operation. The default, two step method is as follows: 1) A baseline is constructed using the Construct Baseline command from the Optimize Menu. 2) The Complete command is selected from the Operation Menu, and the constructed baseline is automatically subtracted from the active scan.

When this check box is not checked, the subtract constructed baseline function becomes a three step operation: 1) A baseline is constructed using the Construct Baseline command from the Optimize Menu. 2) The Complete command is selected from the Operation Menu, the constructed baseline becomes filled in for viewing and is not subtracted from the active scan until; 3) the Sub Constructed Baseline command from the Optimize Menu is selected. By using the three step method you are able to view the constructed baseline in its unsegmented form.

Cross Hair Cursor- When this check box is checked, the mouse cursor becomes a cross hair cursor. This is the default option. If the check box is not checked, the cursor becomes an arrow. This option is strictly a matter of personal preference.

To change any of the Graph Setup . . . parameters, simply click on the appropriate radio button or check box as desired. Click on the Save command and dialog box will close, and the graph will be updated to reflect the changes.

Right Mouse Button Commands

When a scan is displayed in the DISPLAY function, the right mouse button can be pressed to activate an additional command menu. This small menu box contains commands which act upon the active scan, and will only be displayed when the cursor is in the active graph. If more than graph one graph is displayed, the menu box will only be active in the graph containing the active scan. The following are the commands available in this menu box:

Zoom- When the *Zoom* command is activated, the cursor will change to a magnifying glass. To zoom a specific section of the active scan, place the magnifying glass above and to the left of the area you want to box for zooming. Press the left mouse button and drag the mouse down and to the right, until the desired area is boxed. Release the left mouse button. the graph will now update displaying only the area which was boxed. The zoom function is variable, allowing you to zoom in as much as desired. By zooming you are performing a view modification, and any subsequent operations performed on the scan will affect the whole scan, not just the section currently in view. To work only with the zoomed section, click on the *Optimize* Menu and select the *Snip at View* command. This performs a data modification, eliminating all sections of the scan not currently in view.

Scroll- When the *Scroll* command is selected, the cursor will change to a scroll icon, allowing you to drag the scan(s) about the graph. Using *Scroll* command will drag all scans within the graph, and maintain their exact coordinates. This command differs from the drag commands in that the coordinates of the scans are always maintained. When finished scrolling, press the right mouse button and select the *End* command.

End- The *End* command will terminate either the *Zoom* or *Scroll* commands.

Previous- The *Previous* command will toggle between the current display and the previous view modification which was performed. For example, if you zoom in on a section of a scan, then want to look at the full scan again, simply press the right mouse button and select *Previous*. If you wish to go back to the zoomed section simply select *Previous* again. This command will not be operational if a data modification has been performed using the *Snip at View* command.

Restore- If a series of view modifications have been performed, e.g., multiple levels of zoom, and you wish to restore the original scan or the last data modification, simply select the *Restore* command.

Print- The *Print* command allows you to print the active graph and its contents. This is the only way to print the actual scans. When this command is selected, a dialog box will appear stating that the graph is being sent to the printer. To cancel the *Print* command, click on the *Cancel* command button in the dialog box.

Section 3.3 WEIGHT Function

The WEIGHT function of the software will establish a standard curve then determine the molecular weight of unknown proteins by comparing their mobilities to that of the standard curve.

Command Descriptions

MODE Menu

The mode menu is continuous throughout the software program, allowing the user to move rapidly from one function to another without having to go through multiple menu levels. Also available in the *MODE* Menu is the *Exit* command, which allows the user to exit to the main menu to another function.

FUNCTIONS Menu

These commands allow for the entering of protein standard molecular weight values into a table, calculation of a relative mobility vs. log molecular weight standard curve, and calculation of unknown protein molecular weights.

1. FUNCTIONS COMMAND: *Protein Editor*

When this command is selected, the Protein Editor dialog box will appear allowing you either create a new protein molecular weight file (table) or edit an existing one. Each protein name and corresponding molecular weight in a protein molecular weight file is subsequently, associated with a peak in a standard scan, using the *Calculate Standard* function. Note: tables of protein names and molecular weights for the Bio-Rad high and low molecular weight standards are provided with the 1-D program and do not have to be entered.

To create a new protein molecular weight file:

Click on the *FUNCTIONS* menu and select the *Protein Editor* command. A dialog box appears. Enter the names and molecular weight values for your SDS-PAGE Molecular Weight Standards by simply typing in the protein name, pressing [tab], and entering the corresponding weight. Finally click on the [Enter] command button or press [Return]. Continue this procedure until all the standard protein names and molecular weights have been entered. Note that the order of entry is not important as they will automatically be placed in order of decreasing molecular weight. When finished, click on the *Save* command button. A dialog box appears prompting for a filename. Enter a filename and click on the *OK* command button. The filename will automatically be appended with a .MW extension.

To edit an existing protein molecular weight file:

Click on the *FUNCTIONS* menu and select the *Protein Editor* command. A dialog box appears. Click on the *Open* command button. A standard Windows entry box appears listing only the existing protein molecular weight files within the current drive and directory. Double click on the protein file that you wish to open. (For more detailed information on using the standard Windows entry box refer to section 3.1). Once opened, the name of the file will be in the title of the dialog box, and the first protein and corresponding molecular weight of the file will be displayed in the entry boxes.

To add a protein and molecular to the weight table, type over the currently highlighted name and molecular weight and then click on *Enter*. Note: the protein and molecular weight you've typed over is not deleted from the table.

To delete, or change any of the values within the table use the *Next* or *Previous* commands to step through the table until the desired protein and molecular weight are displayed. To delete that table entry press *Delete* and both the protein and corresponding molecular weight will be deleted from the table. To change any protein or molecular weight value for an entry, highlight that value and then enter the new value. Click on *Enter* and the table will be updated to reflect the changes.

To start a new protein molecular weight file:

Click on the *New* command button and a dialog box will appear asking if you want to save changes to the current file. Answer as appropriate. The edit fields will now clear allowing entry of data for a new protein molecular weight file.

2. FUNCTIONS COMMAND: Calculate Standard

The *Calculate Standard* command is used to calculate a relative mobility vs. log molecular weight standard curve from standard scan data. When this command is selected, the following full screen dialog box appears with interior fields: the protein field, the peak field, and the standard field.

Standard Curve Generation (untitled)

Proteins
File: (untitled)

Open...

Peaks
File: (untitled)
Scan: -
Peak Number:

Open...
Assoc.

Standard
Calculate Curve
Save Table

Done

Standard Scan

Protein Field

The protein field contains a scrolling list of proteins and their corresponding molecular weights which can be associated with peaks within a scan file. The *Open* command button in the protein field opens an entry box which prompts for a protein molecular weight filename. This entry box will only list filenames which have the .MW extension. Double click on a protein molecular weight file to open it. To view the complete protein and molecular weight table it may be necessary to use the scroll bar.

Peak Field

The peak field displays a scrolling list of the peaks contained within a standard scanfile which is opened. Clicking on the *Open* command button in the peaks field will display a dialog box which prompts for the name of the file in which the standard proteins scan is stored. Select a standard scan file by double clicking on it. A second dialog box will then appear prompting for the standard scan name. Double click on the desired standard scan name. The table will now list all the peaks which have been previously identified in the scan, i.e. Peak 0 through Peak 5. The scan itself will also be displayed in the graph in the lower part of the screen. The *Assoc.* command button is used to associate a highlighted protein listed in the Protein field, to a appropriate highlighted Peak number in the Peak field. To continue associating proteins and peaks down the tables, simply continue to press the *Associate* command button. Note that the proteins are listed in descending order of molecular weight and that the peaks are listed in descending order of relative mobility.

Standard Field

The standard field contains the command buttons associated with calculation of the standard curve. Once the peaks have been associated with the correct standard protein name and weight, the *Calculate* command button will become highlighted. Click on the *Calculate* command button to specify any or all the Regression types which you would like to have calculated. After clicking on the desired Regression type click on the *OK* command button. The regressions will then be quickly calculated, the Regression type dialog box will disappear, and the *Calculate* button will be greyed. To see the relative mobility vs. log (MW) scatter plot and the different regression curves calculated, click on the *Curve* command button. A scatter plot graph of relative mobility vs. log (MW) will then appear. To display the different regression curves on the graph, simply click on any or all of the check boxes along the bottom. To remove any of the curves, simply click again on the appropriate check box. The graph will update showing the selected regression curves in different colors. This feature is useful for determining which regression type provides the best fit of the data points. Due to the sigmoidal shape of an SDS-PAGE standard curve, the third order regression will often give the best fit of the curve. The user is however cautioned when using higher orders of regression (Appendix D). To close the scatter plot double click on the box in the upper left corner. Once the standard curves have been calculated, the *Save* command button must be selected in order to use this data for calculation of an unknown. The *Table* command button will display a list of the associations. To quickly close the table double click in the upper left hand corner on the menu bar.

Standard Scan Graph

The standard scan which is selected for generation of the standard curve is displayed in this field. The peaks which have been identified in this scan are shaded and have a peak number located above it. If the number of peaks which have been identified in the scan is greater than the number of proteins listed in the molecular weight file, then either the molecular weight file or the scan will have to be edited.

When the cursor is moved into the scan graph the cursor becomes a vertical line which when moved left and right in the graph will correspond to the X (mm) readout value on the bottom of the graph.

3. FUNCTIONS COMMAND: *Calculate Unknown*

The *Calculate Unknown* command allows for the calculation of unknown protein molecular weights using a standard curve generated with the *Calculate Standard* command. Both the standard scan(s) and unknown scan(s) must be contained within the same file to be available for use in the *Calculate Unknown* command. When this command is selected, the following full screen dialog box appears with interior fields: standards, unknowns, unknown scan, and estimated molecular weight.

Calculate Unknown Molecular Weight

File:

Standards

Unknowns

Linear Quad Cubic

Open Table Calculate

Unknown Scan

Linear Quadratic Cubic

Estimated MW

Save... Est. MW Done

Standards Field

The standards field will display all the standard scan names within the opened file, which are available for use in calculation of the unknown molecular weights. Only standard scans which have had peaks identified and a standard curve calculated will be listed in this box. When a standard scan is highlighted in the list box, the regression radio button(s) on the right will be highlighted identifying the types of regressions which have been calculated using this scan. Use the scroll bar if necessary to highlight a desired standard scan.

Unknowns Field

The unknowns field will display all the unknown scan names within the opened file, which are available for use in calculation of unknown molecular weights. All scan names within this list box will have an appendix, e.g. sample.000-a, as only scans which have had peaks identified will be available for molecular weight calculation. Use the scroll bar if necessary to highlight a desired unknown scan. Note that the unknown scan(s) must be from the same scanfile as the standard scan(s).

Unknown Scan Field

The unknown scan field displays a graph of the unknown scan and provides an on-screen readout of the X (mm) location, relative mobility (Rf), and the calculated molecular weights, at the position of the cursor. When the cursor is moved into the graph it changes to a vertical line, enabling you to precisely position the mouse. The default on-screen readout values are for the cubic regression curve type. To change the on-screen readout values to reflect the linear or quadratic regression curves, simply click on the appropriate radio button below the graph.

Estimated MW Field

The estimated MW field displays a table of the relative mobility (Rf), calculated molecular weights (linear, quadratic, and cubic), for each peak identified within the unknown scan. When the *Table* command button is pressed, Rf, MW, and protein names for each peak of the standard scan are displayed in this field. To view the estimated molecular weights again, click on the *Est. MW* command button to the right of the field. Use the scroll bar if necessary to view the full list.

Calculating Unknown Molecular Weight-

To calculate the molecular weight of an unknown, click on the *Open* command button. The Open file entry box will be displayed, listing all scanfiles within the current drive and directory. Double click on the scanfile containing the standard and unknown scan(s) you wish to use in the calculation. Note: the standard and unknown scan(s) must be contained within the same file. Once opened, the file name will be displayed at the top of the dialog box, and the standard and unknown scans available for calculation will be displayed in their respective fields. Scrolling if necessary, highlight both the standard scan and the unknown scan, which you wish to use for the molecular weight calculation. Click on the *Calculate* command button and the unknown scan graph will be displayed, along with the estimated molecular weights. Move the cursor into the graph, and read the relative mobility and molecular weight values for any position along the scan. Click on the linear or quadratic radio buttons so that the readout values reflect those regression curves. To save the calculated molecular weights for the unknown scan click on the *Save* command button before exiting using the *Done* command button. To calculate the molecular weight of another unknown scan in the opened scanfile, using the same standard scan, highlight the new unknown scan name then press *Calculate*. The new unknown scan graph will be displayed and the estimated molecular weight table will be updated to reflect the new scan. If calculation of unknown using another standard scan is desired, simply highlight the unknown scan and the new standard scan then press *Calculate*.

Section 3.4 REPORTS Function

The REPORTS Function allows you to print a report on any scan contained within any file. The reports include: densitometer settings prior to export of the scan, information about the scan itself, a peak report if peaks have been identified, and a molecular weight report if molecular weights have been calculated. Note: to print the scan in a graph, the scan must be displayed using the DISPLAY function and printed by pressing the right mouse button and selecting the *Print* command.

Command Descriptions

FUNCTIONS Menu

These commands allow for the selection of the report contents and the scan report which you wish to print.

1. FUNCTIONS COMMAND: Contents

When this command is selected, a dialog box appears containing check boxes for each section of a report which can be printed. Each report will always contain the

scan name, scan type, the file name where it exists, and the date when it was printed. The following is a description of the report sections which can be printed:

Densitometer Data- When this check box is checked, a list of the operations which have been performed by the densitometer will be printed. For example: if a scan was filtered on the densitometer before exporting to the 1-D software, the user would have identified the scan as having been filtered by entering the filtering value in the *CAPTURE* function, before capturing the scan. These pre-capture scan optimization values will be reported in this section.

Scan Data- When this check box is checked, a section of general scan information will be printed. This section will include the origin and front X (mm) location, the number of peaks identified in the scan, and the total area under the scan curve.

Peak Report- When this check box is checked, information about each identified peak will be printed in a table. This information includes: identification of whether Gaussian modeling has been performed, peak number, position within the scan, distance from the origin, peak relative mobility, maximum peak height, peak width, percentage of total area peak comprises, and amount of material in peak. Amount of material will only be calculated if track loading (amount applied to gel) was specified when the scan was captured from the densitometer.

Molecular Weight Report- When this check box is checked, the molecular weight report section will be printed. If the scan being printed is a standard scan then the report will be a standard report consisting of: the peak number, peak name, relative mobility (Rf), and known molecular weight. Additionally, information regarding the standard curves will be provided. This includes: the equation, the coefficient values, R squared, mean Rf, mean Rf standard error, for each order of regression. This information is used to determine the regression order which provides the best fit, or most useful information. Normally, the regression order providing the highest R squared value, and the lowest standard error would provide the most accurate molecular weight values. However the user is cautioned when using higher orders of regression (see Appendix D).

If the scan is an unknown scan then the report will consist of: the unknown peak number, its relative mobility, and the molecular weights as calculated using the first, second, and third order regressions.

Left Printer Margin- To adjust the left printer margin simply enter a value here between 0 and 8.

After selecting the desired sections you wish to have printed for a particular scan click on the *Set* command button.

2. FUNCTIONS COMMAND: Print

When this command is selected, the open file entry box will appear. Double click on the file which contains the scan report(s) you wish to print. A new dialog box will appear containing a list box of the scans contained within the scanfile you have opened. Click on a scan name to highlight it. The two fields to the right contain radio buttons which will highlight, identifying the type of scan and any operations which have been performed on it. Click on the *Add* command button to send this scan report to the printer.

Appendices

APPENDIX A: TECHNICAL SUPPORT

Bio-Rad provides complete technical support on all features of the 1-D Analyst II software. If you have any problems, or any questions on the use of the features, please call Bio-Rad Technical Service at:

1-800-4-BIO-RAD (1-800-424-6723)

Please report any reproducible software problems to Bio-Rad Technical Service so we can eliminate them from future versions.

APPENDIX B: CABLE PIN-OUT TABLES

9-PIN SERIAL PORT

Video Densitometer serial interface cable to serial/parallel interface card (9-pin serial port) - equivalent to Catalog Number 165-2057:

DENSITOMETER

(Male 25-pin) (Female 9-pin)

<u>Signal</u>	<u>Pin #</u>
RxD	2
TxD	3
RTS	4
CTS	5
DSR	6
GND	7
DTR	20

IBM PC

<u>Pin #</u>	<u>Signal</u>
3	TxD
2	RxD
8	CTS
7	RTS
4	DTR
5	GND
6	DSR

25-PIN SERIAL PORT

Video Densitometer serial interface cable to asynchronous communications port (25-pin serial port) :

DENSITOMETER

(Male 25-pin)

Signal	Pin.#
RxD	2
TxD	3
RTS	4
CTS	5
DSR	6
GND	7
DTR	20

IBM PC

(Female 25-pin)

Pin.#	Signal
2	TxD
3	RxD
5	CTS
4	RTS
20	DTR
7	GND
6	DSR

APPENDIX C: Data Files and Data Base Management

When a raw scan is received by 1-D Analyst II software, the data is stored in binary in the following format: scanfile.SCN.

As individual raw unprocessed scans are captured and stored into the scanfiles, they are sequentially given a three digit extension to their root name, starting with .000, in the format: root name.XXX, i.e., Data.000-a.

Scans which are processed in any manner, i.e., background subtracted, peaks identified, etc., and then saved using the *Save* function will be saved in the format: root name.XXX-a.

Molecular weight standards data table files are stored in the following format: standardfile.MW.

Managing the Data Base:

Periodically it will be necessary to delete unwanted scanfiles, created by the 1-D Analyst II software, from your hard disk in order to free-up hard disk space. The scanfiles can be archived on floppy disks for future use. Deletion of scanfiles is accomplished using the MS-DOS EXECUTIVE window, while deletion of the scans contained within the scanfiles is accomplished using the *Delete Scan ...* function from the DISPLAY window.

To delete scanfiles from a hard disk use the following procedure:

(1) Bring up the MS-DOS Executive screen by either double clicking on the disk icon in the lower left corner of the screen, or by clicking once on the minimize box in the upper right corner. (2) Specify which disk drive you wish to delete files from by clicking on the appropriate icon. (3) Point and click on the file which you wish to delete. (4) Click on FILE in the menu bar and then click on the DELETE command. The file will then be deleted from the disk drive. Continue this process until all of the unwanted files are deleted.

To delete unwanted scans from within files, use the following procedure:

(1) After initiating the 1-D Analyst II program, click on the *MODE* menu and select *Display* from the menu. (2) Click on the *SCANFILE* menu and select *Delete Scan ...*

from the menu. (3) Double click on the file name which contains the scan(s) which you wish to delete. A dialog box showing the scans contained within the select file will appear. (4) To delete one scan, double click on the single scan to be deleted. To delete many scans from the selected scanfile, press and hold the shift button and click the left mouse button on all scans which you wish to delete. Click on the *Delete* command button to delete the selected scans. The "Open File Name:" dialog box will then reappear enabling you to delete scans from another scanfile. If you are finished deleting scans, click on the *Exit* command button to go back to the *Display* window.

APPENDIX D: Use of Estimated Molecular Weights

1. The user is cautioned when using the estimated molecular weights calculated with higher order regressions. Higher orders of regression often result in a "better fit" because they make more concessions to particular measurement errors in the data used to derive the standard curve. While this can increase the amount of useful information, it can also reduce the likelihood that the standard curve represents the sample data. This is because the precision with which the standards data are modeled is not always matched by the sample data.

2. The user is also cautioned when using sample data where the x and y values do not fall within the x and y range used to calculate the standard curve. Using these values outside of the standard curve x and y range may result in undefined or absurd values (the 1-D analyst will display 0.000 for values outside of the range). If higher orders of regression of standard data produce several undefined sample values, then it may not be the best model to choose for the standard curve.

APPENDIX E: Troubleshooting

Windows Problems

Problem

Mouse pointer does not appear after booting Microsoft Windows.

Solution

Logitech serial mouse was not specified during the Microsoft Windows setup procedure.

Normally, Microsoft Windows will find the serial mouse whether it is in communications port number one or in number two. Exit Windows by pressing [ALT]+[F4] followed by the return key. Type: *WIN* or *WIN386* again. If the mouse pointer still does not appear, check the appropriate hardware connections and/or consult your Windows User's Guide.

Capture Mode Problems

Problem

An "Error opening port" dialog box, followed by an "Open comm error: MEMORY" dialog box, appear after pressing the CAPTURE command button.

The Capture Screen locks up while transferring data from the densitometer.

Solution

The Microsoft Windows mouse driver included is not compatible with your computer system. The mouse driver supplied with your mouse must be installed and the appropriate changes made to your config.sys file.

The OUTPUT DATA command from the densitometer's Output Data Screen was pressed before the CAPTURE Command was pressed on the 1-D Analyst II software. Reboot the computer and the densitometer if necessary.

Display Problems

Problem

The Automatic Peak Identification function is not identifying all the peaks as expected.

Solution

The threshold values previously set in the Peak Criteria function are too high and should be lowered.

The smoothing window calculated by the 1-D Analyst II software does not consider that region of the scan to be a peak in relation to the rest of the scan. Enter a smoothing window value under Peak Criteria, which is either higher or lower than the value calculated by the program, as appropriate for the section of the scan in question.

An operation or function has no effect on a scan.

The correct criteria for the operation or function has not been set.

The scan in question is not the "Active Scan" in the graph. Double click on the scan to make it active.

Index

- Activating a window 12
- Axis Minimums and Maximums 36
- Background 35
- Background scan 35, 40
- Baseline subtracted 36
- Boost factor 36
- Calculate standard 59
- Calculate unknown 60
- Capture function 31
- Changing windows 11
- Clicking 10
- Closing a window 12
- Command buttons 13
- Command menus 9
- Contents 62
- Data modification 41
- Densitometer data 63
- Densitometer settings 35
- Dialog boxes 12
- Display 37
- Display function terminology 37
- Double-clicking 10
- Dragging 10
- Enhance command
 - Criteria 47
 - Enhance 48
 - Smooth 48
- Enhancement frequency 36
- Entry fields 13
- Error opening port 66
- File command
 - Append 33
 - Create 32
- Filter frequency 36
- Functions menu 58
- Graph setup 55
- High resolution lens 36
- Interpolation command 33
- Left printer margin 63
- Maximize box 9
- Maximizing a window 11

- Menu bar 9
- Microsoft Windows 4, 6, 8
- Minimize box 9
- Minimizing a window 11
- Mode menu 38, 58
- Molecular weight report 63
- Mouse pointer 9
- Operations command
 - Cancel 54
 - Complete 53
- Optimize command
 - Align 2 pts 45
 - Align 4 pts 45
 - Construct baseline 43
 - Difference scan 44
 - DragX 46
 - DragXY 46
 - DragY 46
 - Flip 45
 - Front 45
 - Origin 44
 - Sub construct baseline 43
 - Sub horizontal baseline 43
 - Subtract background 44
- Peak report 63
- Peaks command
 - Auto identify 51
 - Edit 50
 - Gaussian 51
 - Integrate interactive 52
 - Peak criteria . . . 49
 - Peak info 52
- Pointing 10
- Press 10
- Print 63
- Protein editor 58
- Radio buttons 12
- Reports function 62
- Scan command
 - Add . . . 41
 - Close graph 41
 - Delete 41
 - New . . . 39

- Original 41
- Previous 42
- Save 42
- Save as . . . 42
- Scan info 42
- Scan data 63
- scan information box 42
- Scan parameters
 - Root name entry field 35
 - Type 35
- Scanfile 38
- Scanfile command
 - Delete Scan . . . 38
 - Compact Scanfile 39
- Scroll bars 10
- Scrolling within a window 11
- Serial mouse 5
- Setup command
 - Colors 54
- Sizing a window 11
- Software functions 31
- Standard 35
- Standard scan 35, 40
- System menu box 9
- Text box 13
- Transfer Command
 - Capture 34
 - Set comm 34
 - Stop 34
- Tutorial summary 28
- Unknown scan 35, 41
- Unknowns 35, 41
- View modification 42
- Weight function 58
- Work area 9

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